arXiv:2102.03334v2 [stat.ML] 10 Jun 2021

# ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision

**Wonjae Kim** [*1 †] **Bokyung Son** [*1] **Ildoo Kim** [2]


## Abstract

Vision-and-Language Pre-training (VLP) has improved performance on various joint vision-and-language downstream tasks. Current approaches to VLP heavily rely on image feature extraction processes, most of which involve region supervision (e.g., object detection) and the convolutional architecture (e.g., ResNet). Although disregarded in the literature, we find it problematic in terms of both (1) efficiency/speed, that simply extracting input features requires much more computation than the multimodal interaction steps; and (2) expressive power, as it is upper bounded to the expressive power of the visual embedder and its predefined visual vocabulary. In this paper, we present a minimal VLP model, Vision-and-Language Transformer (ViLT), monolithic in the sense that the processing of visual inputs is drastically simplified to just the same convolution-free manner that we process textual inputs. We show that ViLT is up to tens of times faster than previous VLP models, yet with competitive or better downstream task performance. Our code and pre-trained weights are available at https://github.com/dandelin/vilt.

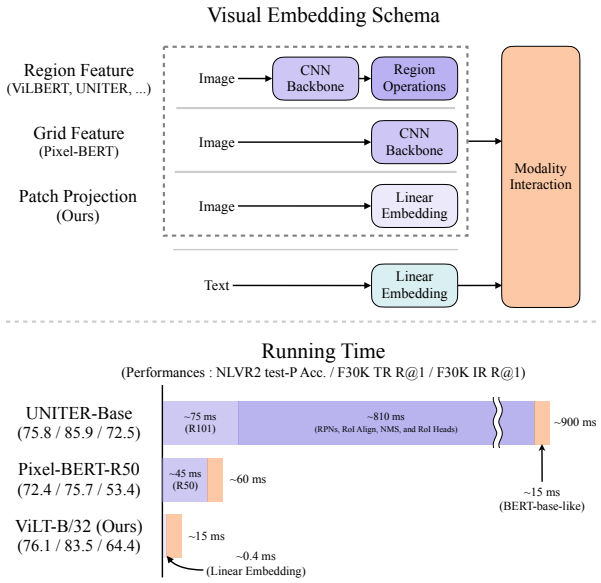




*Figure 1.* Visual comparison of conventional VLP architectures and our proposed ViLT. We have entirely removed convolutional neural networks from the VLP pipeline without hurting performance on downstream tasks. ViLT is the first VLP model of which the modal-specific components require *less* computation than the transformer component for multimodal interactions.

## 1. Introduction

The pre-train-and-fine-tune scheme has been expanded to a joint domain of vision and language, giving birth to the category of *Vision-and-Language Pre-training (VLP)* models (Lu et al., 2019; Chen et al., 2019; Su et al., 2019; Li et al., 2019; Tan & Bansal, 2019; Li et al., 2020a; Lu et al., 2020; Cho et al., 2020; Qi et al., 2020; Zhou et al., 2020; Huang et al., 2020; Li et al., 2020b; Gan et al., 2020; Yu et al., 2020; Zhang et al., 2021). These models are pre-trained with image text matching and masked language modeling objectives[1] on images and their aligned descriptions, and are fine-tuned on vision-and-language downstream tasks where the inputs involve two modalities.

To be fed into VLP models, image pixels need to be initially embedded in a dense form alongside language tokens. Since the seminal work of Krizhevsky et al. (2012), deep convolutional networks have been regarded as essential for this visual embedding step. Most VLP models employ an object detector pre-trained on the Visual Genome dataset (Krishna et al., 2017) annotated with 1,600 object classes and 400 attribute classes as in Anderson et al. (2018). Pixel-

[*]Equal contribution [†]Current affiliation: NAVER AI Lab, Seongnam, Gyeonggi, Republic of Korea. [1]Kakao Enterprise, Seongnam, Gyeonggi, Republic of Korea [2]Kakao Brain, Seongnam, Gyeonggi, Republic of Korea. Correspondence to: Wonjae Kim <wonjae.kim@navercorp.com>.

*Proceedings of the $38^{th}$ International Conference on Machine Learning*, PMLR 139, 2021. Copyright 2021 by the author(s).

[1]While some works employ additional objectives and data structures, these two objectives apply to almost every VLP model.

1202 nuJ 01 ]LM .tats[ 2v43330.2012:viXra

# ViLT：无卷积或区域监督的视觉-语言 Transformer

金元宰 *1† 孙宝京 *1 金一斗 2


## 摘要

视觉-语言预训练（VLP）提升了各种视觉与语言联合下游任务的性能。当前的 VLP 方法在很大程度上依赖图像特征提取过程，其中大多数涉及区域监督（例如目标检测）和卷积架构（例如 ResNet）。尽管这一点在文献中常被忽视，但我们发现它在以下两个方面存在问题：（1）效率/速度：仅提取输入特征所需的计算量就远高于多模态交互步骤；（2）表达能力：其上限受限于视觉嵌入器及其预定义视觉词表的表达能力。本文提出了一种极简的 VLP 模型——视觉-语言 Transformer（ViLT）。所谓整体式，是指视觉输入的处理被大幅简化为与文本输入处理相同的无卷积方式。我们表明，ViLT 的速度比以往的 VLP 模型快数十倍，同时在下游任务性能上仍具有竞争力甚至更优。我们的代码和预训练权重可在 https://github.com/dandelin/vilt 获取。

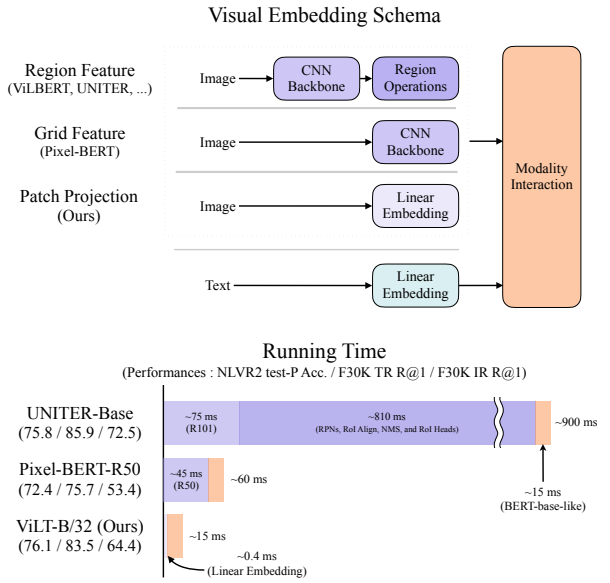




图1. 传统 VLP 架构与我们提出的 ViLT 的视觉比较。我们已将卷积神经网络完全从 VLP 流水线中移除，同时不损害下游任务的性能。ViLT 是首个其模态特定组件所需的计算量比用于多模态交互的 transformer 组件 *less* 的 VLP 模型。

## 1. 引言

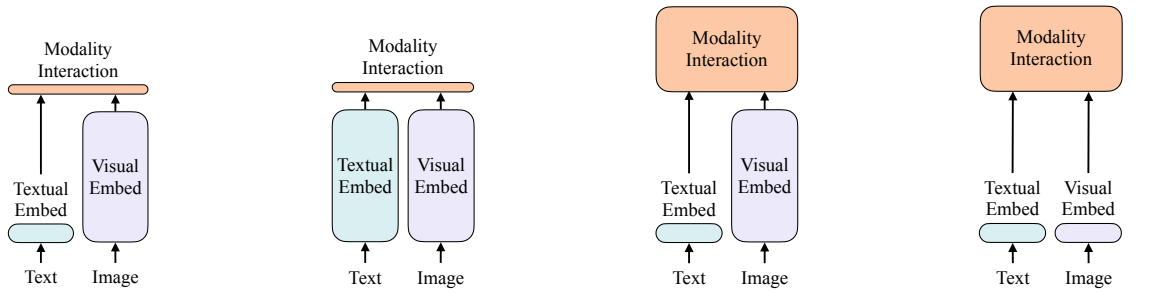
预训练—微调方案已扩展到视觉与语言的联合领域，催生了*Vision-and-Language Pre-training (VLP)*模型这一类别（Lu 等，2019；Chen 等，2019；Su 等，2019；Li 等，2019；Tan 和 Bansal，2019；Li 等，2020a；Lu 等，2020；Cho 等，2020；Qi 等，2020；Zhou 等，2020；Huang 等，2020；Li 等，2020b；Gan 等，2020；Yu 等，2020；Zhang 等，2021）。这些模型在图像及其对齐的描述上，以图像-文本匹配和掩码语言建模目标[1]进行预训练，并在输入涉及两种模态的视觉-语言下游任务上进行微调。

为了输入到 VLP 模型中，图像像素需要首先与语言标记一起以密集形式嵌入。自 Krizhevsky 等人（2012）的开创性工作以来，深度卷积网络一直被认为是这一视觉嵌入步骤必不可少的。大多数 VLP 模型采用在 Visual Genome 数据集（Krishna 等人，2017）上预训练的目标检测器，该数据集如 Anderson 等人（2018）中所示，标注了 1,600 个对象类别和 400 个属性类别。像素-

*Equal contribution †Current affiliation: NAVER AI Lab, Seongnam, Gyeonggi, Republic of Korea. [1]Kakao Enterprise, Seongnam, Gyeonggi, Republic of Korea [2]Kakao Brain, Seongnam, Gyeonggi, Republic of Korea. Correspondence to: Wonjae Kim <wonjae.kim@navercorp.com>.

*Proceedings of the $38^{th}$ International Conference on Machine Learning*, PMLR 139, 2021. Copyright 2021 by the author(s).

1虽然一些工作采用了额外的目标和数据结构，但这两个目标几乎适用于每个VLP模型。




(a) VE > TE > MI (b) VE = TE > MI (c) VE > MI > TE (d) MI > VE = TE

*Figure 2.* Four categories of vision-and-language models. The height of each rectangle denotes its relative computational size. VE, TE, and MI are short for visual embedder, textual embedder, and modality interaction, respectively.

BERT (Huang et al., 2020) is one exception of this trend, as it uses ResNet variants (He et al., 2016; Xie et al., 2017) pre-trained on ImageNet classification (Russakovsky et al., 2015) embedding pixels in place of object detection modules.

To this date, most VLP studies have focused on improving performance by increasing the power of visual embedders. The shortcomings of having a heavy visual embedder are often disregarded in academic experiments because region features are commonly cached in advance at training time to ease the burden of feature extraction. However, the limitations are still evident in real-world applications as the queries in the wild have to undergo a slow extraction process.

To this end, we shift our attention to the lightweight and fast embedding of visual inputs. Recent work (Dosovitskiy et al., 2020; Touvron et al., 2020) demonstrated that using a simple linear projection of a patch is effective enough to embed pixels before feeding them into transformers. Whereas being the solid mainstream for text (Devlin et al., 2019), it is only recently that transformers (Vaswani et al., 2017) are used for images as well. We presume that the transformer module– used for modality interaction in VLP models– can also manage to process visual features in place of a convolutional visual embedder, just as it processes textual features.

This paper proposes the Vision-and-Language Transformer (ViLT) that handles two modalities in a single unified manner. It mainly differs from previous VLP models in its shallow, convolution-free embedding of pixel-level inputs. Removing deep embedders solely dedicated to visual inputs significantly cuts down the model size and running time by design. Figure 1 shows that our parameter-efficient model is tens of times faster than VLP models with region features and at least four times faster than those with grid features while exhibiting similar or even better performance on vision-and-language downstream tasks.

Our key contributions can be summarized as follows:

- ViLT is the *simplest* architecture by far for a vision-and-language model as it commissions the transformer module to extract and process visual features in place of a separate deep visual embedder. This design inherently leads to significant runtime and parameter efficiency.
- For the first time, we achieve competent performance on vision-and-language tasks without using region features or deep convolutional visual embedders in general.
- Also, for the first time, we empirically show that whole word masking and image augmentations that were unprecedented in VLP training schemes further drive downstream performance.

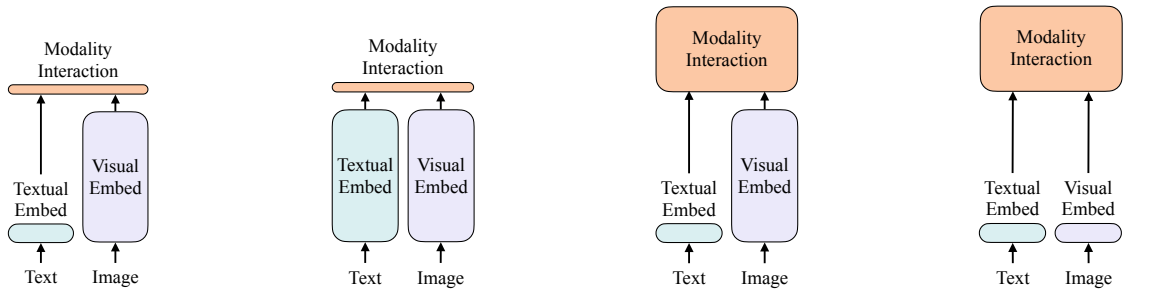
## 2. Background

### 2.1. Taxonomy of Vision-and-Language Models

We propose a taxonomy of vision-and-language models based on two points: (1) whether the two modalities have an even level of expressiveness in terms of dedicated parameters and/or computation; and (2) whether the two modalities interact in a deep network. A combination of these points leads to four archetypes in Figure 2.

The *visual semantic embedding* (VSE) models such as VSE++ (Faghri et al., 2017) and SCAN (Lee et al., 2018) belong to Figure 2a. They use separate embedders for image and text, with the former being much heavier. Then, they represent the similarity of the embedded features from the two modalities with simple dot products or shallow attention layers.

CLIP (Radford et al., 2021) belongs to Figure 2b as it uses separate but equally expensive transformer embedders for each modality. Interaction between the pooled image vector and text vector is still shallow (dot product). Despite CLIP's remarkable zero-shot performance on image-to-text




(a) VE > TE > MI (b) VE = TE > MI (c) VE > MI > TE (d) MI > VE = TE

图 2. 四类视觉与语言模型。每个矩形的高度表示其相对计算规模。VE、TE 和 MI 分别是视觉嵌入器、文本嵌入器和模态交互的缩写。

BERT（Huang 等，2020）是这一趋势的一个例外，因为它使用在 ImageNet 分类（Russakovsky 等，2015）上预训练的 ResNet 变体（He 等，2016；Xie 等，2017）来嵌入像素，以替代目标检测模块。

迄今为止，大多数 VLP 研究都专注于通过增强视觉嵌入器的能力来提升性能。拥有笨重视觉嵌入器的缺点在学术实验中往往被忽视，因为在训练时通常会提前缓存区域特征，以减轻特征提取的负担。然而，在真实应用中，这些局限性仍然十分明显，因为实际场景中的查询必须经过缓慢的提取过程。

为此，我们将注意力转向视觉输入的轻量级、快速嵌入。近期工作（Dosovitskiy et al., 2020；Touvron et al., 2020）表明，在将像素输入 Transformer 之前，使用简单的图像块线性投影已足以有效地嵌入像素。尽管 Transformer（Vaswani et al., 2017）在文本领域已成为稳固的主流（Devlin et al., 2019），但直到最近才被用于图像。我们推测，在 VLP 模型中用于模态交互的 Transformer 模块，也能够替代卷积式视觉嵌入器来处理视觉特征，就像它处理文本特征一样。

本文提出了视觉-语言 Transformer（ViLT），它以单一统一的方式处理两种模态。它与以往 VLP 模型的主要区别在于，其对像素级输入采用了浅层、无卷积的嵌入方式。去除专门用于视觉输入的深层嵌入器，从设计上显著减少了模型规模和运行时间。图 1 显示，我们这种参数高效的模型比使用区域特征的 VLP 模型快数十倍，并且比使用网格特征的模型至少快四倍，同时在视觉-语言下游任务上表现出相近甚至更好的性能。

我们的主要贡献可概括如下：

- ViLT 是迄今为止视觉-语言模型中最 *simplest* 的架构，因为它让 Transformer 模块代替单独的深度视觉嵌入器来提取和处理视觉特征。这种设计本质上带来了显著的运行时效率和参数效率提升。
- 我们首次在不使用区域特征或深度卷积视觉嵌入器的情况下，在视觉与语言任务上取得了有竞争力的性能。
- 此外，我们首次通过实验证明，在 VLP 训练方案中前所未有的全词掩码和图像增强能够进一步提升下游性能。

## 2. 背景

### 2.1. 视觉与语言模型的分类体系

我们基于两点提出一种视觉-语言模型的分类法：（1）两种模态在专用参数和/或计算方面是否具有均衡的表达能力；以及（2）两种模态是否在深度网络中进行交互。这两点的组合形成了图 2 中的四种原型。

*visual semantic embedding* (VSE) 模型，如 VSE++ (Faghri 等，2017) 和 SCAN（Lee 等，2018），属于图 2a。它们分别使用图像和文本嵌入器，其中前者要重得多。然后，它们通过简单的点积或浅层注意力层来表示来自两种模态的嵌入特征之间的相似性。

CLIP（Radford 等，2021）属于图 2b，因为它为每种模态使用了独立但成本同样高昂的 Transformer 嵌入器。池化后的图像向量与文本向量之间的交互仍然较浅（点积）。尽管 CLIP 在图像到文本的零样本性能方面表现出色

retrieval, we could not observe the same level of performance on other vision-and-language downstream tasks. For instance, fine-tuning the MLP head on NLVR2 (Suhr et al., 2018) with the dot product of pooled visual and textual vectors from CLIP as the multimodal representation gives a low dev accuracy of $50.99 \pm 0.38$ (ran with three different seeds); as chance level accuracy is 0.5, we conclude that the representations are incapable of learning this task. It also matches the findings of Suhr et al. (2018) that all models with simply fused multimodal representation failed to learn NLVR2.

This result backs up our speculation that simple fusion of outputs even from high-performing unimodal embedders may not be sufficient to learn complex vision-and-language tasks, bolstering the need for a more rigorous inter-modal interaction scheme.

Unlike models with shallow interaction, the more recent VLP models that fall under Figure 2c use a deep transformer to model the interaction of image and text features. Aside from the interaction module, however, convolutional networks are still involved in extracting and embedding image features, which accounts for most of the computation as depicted in Figure 1. Modulation-based vision-and-language models (Perez et al., 2018; Nguyen et al., 2020) also fall under Figure 2c, with their visual CNN stems corresponding to visual embedder, RNNs producing the modulation parameters to textual embedder, and modulated CNNs to modality interaction.

Our proposed ViLT is the first model of type Figure 2d where the embedding layers of raw pixels are shallow and computationally light as of text tokens. This architecture thereby concentrates most of the computation on modeling modality interactions.

### 2.2. Modality Interaction Schema

At the very core of contemporary VLP models lie transformers. They get visual and textual embedding sequences as input, model inter-modal and optionally intra-modal interactions throughout layers, then output a contextualized feature sequence.

Bugliarello et al. (2020) classifies interaction schema into two categories: (1) *single-stream* approaches (e.g., VisualBERT (Li et al., 2019), UNITER (Chen et al., 2019)) where layers collectively operate on a concatenation of image and text inputs; and (2) *dual-stream* approaches (e.g., ViLBERT (Lu et al., 2019), LXMERT (Tan & Bansal, 2019)) where the two modalities are not concatenated at the input level. We follow the single-stream approach for our interaction transformer module because the dual-stream approach introduces additional parameters.

### 2.3. Visual Embedding Schema

Whereas all performant VLP models share the same textual embedder– tokenizer from pre-trained BERT, word and position embeddings resembling those of BERT– they differ on visual embedders. Still, in most (if not all) cases, visual embedding is the bottleneck of existing VLP models. We focus on cutting corners on this step by introducing patch projection instead of using region or grid features for which heavy extraction modules are used.

**Region Feature.** VLP models dominantly utilize region features, also known as bottom-up features (Anderson et al., 2018). They are obtained from an off-the-shelf object detector like Faster R-CNN (Ren et al., 2016).

The general pipeline of generating region features is as follows. First, a region proposal network (RPN) proposes regions of interest (RoI) based on the grid features pooled from the CNN backbone. Non-maximum suppression (NMS) then reduces the number of RoIs to a few thousand. After being pooled by operations such as RoI Align (He et al., 2017), the RoIs go through RoI heads and become region features. NMS is again applied to every class, finally reducing the number of features under a hundred.

The above process involves several factors that affect the performance and runtime: the backbone, the style of NMS, the RoI heads. Previous works were lenient with controlling these factors, making varying choices from each other as listed in Table 7.[2]

- Backbone: ResNet-101 (Lu et al., 2019; Tan & Bansal, 2019; Su et al., 2019) and ResNext-152 (Li et al., 2019; 2020a; Zhang et al., 2021) are two commonly used backbones.
- NMS: NMS is typically done in a *per-class* fashion. Applying NMS to each and every class becomes a major runtime bottleneck with a large number of classes, *e.g.* 1.6K in the VG dataset (Jiang et al., 2020). *Class-agnostic* NMS was recently introduced to tackle this issue (Zhang et al., 2021).
- RoI head: C4 heads were initially used (Anderson et al., 2018). FPN-MLP heads were introduced later (Jiang et al., 2018). As heads operate for each and every RoI, they pose a substantial runtime burden.

However lightweight, object detectors are less likely to be faster than the backbone or a single-layer convolution. Freezing the visual backbone and caching the region features in advance only helps at training time and not during

[2] Bugliarello et al. (2020) showed that a controlled setup bridges the performance gap of various region-feature-based VLP models.

检索方面，我们未能在其他视觉与语言下游任务上观察到同等水平的性能。例如，在 NLVR2（Suhr 等，2018）上微调 MLP 头部，并将来自 CLIP 的池化视觉向量和文本向量的点积作为多模态表示，得到的开发集准确率较低，为 $50.99 \pm 0.38$（使用三个不同随机种子运行）；由于随机水平的准确率为 0.5，我们认为这些表示无法学习该任务。这也与 Suhr 等（2018）的发现一致，即所有仅采用简单融合多模态表示的模型都无法学习 NLVR2。

这一结果支持了我们的推测：即使是来自高性能单模态嵌入器的输出，简单融合也可能不足以学习复杂的视觉-语言任务，从而进一步凸显了对更严谨的模态间交互方案的需求。

与浅层交互模型不同，图 2c 所示的较新 VLP 模型使用深层 Transformer 来建模图像与文本特征之间的交互。然而，除交互模块之外，卷积网络仍然参与图像特征的提取和嵌入，而这占据了大部分计算量，如图 1 所示。基于调制的视觉-语言模型（Perez et al., 2018; Nguyen et al., 2020）也属于图 2c 的范畴，其中视觉 CNN 主干对应于视觉嵌入器，RNN 生成调制参数并对应于文本嵌入器，而调制后的 CNN 则对应于模态交互。

我们提出的 ViLT 是首个如图 2d 所示类型的模型，其中原始像素的嵌入层与文本标记一样浅且计算量轻。因此，该架构将大部分计算集中在建模模态交互上。

### 2.2. 模态交互模式

当代 VLP 模型的核心是 Transformer。它们以视觉和文本嵌入序列作为输入，在各层中建模跨模态以及可选的模态内交互，然后输出上下文化的特征序列。

Bugliarello 等人（2020）将交互模式分为两类：（1）*single-stream* 方法（例如 Visual-BERT（Li 等人，2019）、UNITER（Chen 等人，2019）），其中各层共同作用于图像和文本输入的拼接表示；以及（2）*dual-stream* 方法（例如 ViLBERT（Lu 等人，2019）、LXMERT（Tan 和 Bansal，2019）），其中两种模态在输入层面并未拼接。我们在交互 Transformer 模块中采用单流方法，因为双流方法会引入额外参数。

### 2.3. 视觉嵌入模式

尽管所有高性能的 VLP 模型都共享相同的文本嵌入器——来自预训练 BERT 的分词器，以及类似于 BERT 的词嵌入和位置嵌入——但它们在视觉嵌入器上有所不同。然而，在大多数（如果不是全部）情况下，视觉嵌入都是现有 VLP 模型的瓶颈。我们专注于在这一步骤上简化流程，引入补丁投影，而不是使用需要繁重提取模块的区域特征或网格特征。

区域特征。VLP 模型主要利用区域特征，也称为自底向上特征（Anderson 等，2018）。它们是从现成的目标检测器（如 Faster R-CNN（Ren 等，2016））中获得的。

生成区域特征的一般流程如下。首先，区域候选网络（RPN）基于从 CNN 骨干网络汇聚得到的网格特征提出感兴趣区域（RoI）。随后，非极大值抑制（NMS）将 RoI 的数量减少到几千个。在经过 RoI Align（He et al., 2017）等操作进行池化后，RoI 通过 RoI 头部并转化为区域特征。随后再次对每个类别应用 NMS，最终将特征数量减少到一百个以内。

上述过程涉及多个影响性能和运行时间的因素：骨干网络、NMS 的风格以及 RoI 头。以往工作在控制这些因素方面较为宽松，彼此之间做出了不同的选择，如表 7 所示。[2]

- 骨干网络：ResNet-101（Lu et al., 2019；Tan & Bansal, 2019；Su et al., 2019）和 ResNext-152（Li et al., 2019；2020a；Zhang et al., 2021）是两种常用的骨干网络。
- NMS：NMS 通常以 *per-class* 方式执行。对于大量类别，将 NMS 应用于每一个类别会成为主要的运行时瓶颈，*e.g.* 在 VG 数据集中有 1.6K 个类别（Jiang et al., 2020）。最近引入了 *Class- agnostic* NMS 来解决这一问题（Zhang et al., 2021）。
- RoI 头：最初使用的是 C4 头（Anderson 等，2018）。后来引入了 FPN-MLP 头（Jiang 等，2018）。由于头部会对每一个 RoI 进行操作，因此会带来相当大的运行时负担。

然而，即使再轻量，对象检测器也不太可能比骨干网络或单层卷积更快。冻结视觉骨干网络并提前缓存区域特征只在训练时有帮助，而在推理期间则无济于事。

2Bugliarello et al. (2020)
表明，受控设置弥合了各种基于区域特征的VLP模型之间的性能差距。

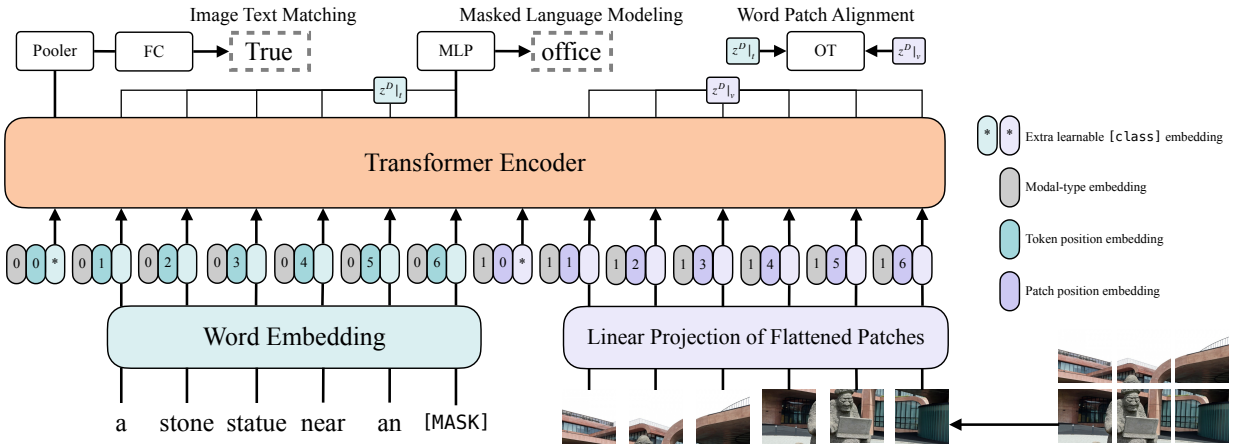



*Figure 3.* Model overview. Illustration inspired by Dosovitskiy et al. (2020).

inference, not to mention that it could hold performance back.

**Grid Feature.** Besides detector heads, the output feature grid of convolutional neural networks such as ResNets can also be used as visual features for vision-and-language pre-training. Direct use of grid features was first proposed by VQA-specific models (Jiang et al., 2020; Nguyen et al., 2020), mainly to avoid using severely slow region selection operations.

X-LXMERT (Cho et al., 2020) revisited grid features by fixing the region proposals to grids instead of those from the region proposal networks. However, their caching of features excluded further tuning of the backbone.

Pixel-BERT is the only VLP model that replaces the VG-pre-trained object detector with a ResNet variant backbone pre-trained with ImageNet classification. Unlike frozen detectors in region-feature-based VLP models, the backbone of Pixel-BERT is tuned during vision-and-language pre-training. The downstream performance of Pixel-BERT with ResNet-50 falls below region-feature-based VLP models, but it matches that of other competitors with the use of a much heavier ResNeXt-152.

We claim that grid features are not the go-to option, however, since deep CNNs are still expensive that they account for a large portion of the whole computation as in Figure 1.

**Patch Projection.** To minimize overhead, we adopt the simplest visual embedding scheme: *linear projection* that operates on image patches. The patch projection embedding was introduced by ViT (Dosovitskiy et al., 2020) for image classification tasks. Patch projection drastically simplifies the visual embedding step to the level of textual embedding, which also consists of simple projection (lookup) operations. We use a $32 \times 32$ patch projection which only requires 2.4M parameters. This is in sharp contrast to complex ResNe(X)t backbones[3] and detection components. Its running time is also ignorable as shown in Figure 1. We make a detailed runtime analysis in Section 4.6.

## 3. Vision-and-Language Transformer

### 3.1. Model Overview

ViLT has a succinct architecture as a VLP model with a minimal visual embedding pipeline and following the single-stream approach.

We deviate from the literature that we initialize the interaction transformer weights from pre-trained ViT instead of BERT. Such initialization exploits the power of the interaction layers to process visual features while lacking a separate deep visual embedder. [4]

$$\bar{t} = [t_{\text{class}}; t_1 T; \cdots; t_L T] + T^{\text{pos}} \quad (1)$$

$$\bar{v} = [v_{\text{class}}; v_1 V; \cdots; v_N V] + V^{\text{pos}} \quad (2)$$

$$z^0 = [\bar{t} + t^{\text{type}}; \bar{v} + v^{\text{type}}] \quad (3)$$

$$\hat{z}^d = \text{MSA}(\text{LN}(z^{d-1})) + z^{d-1}, \qquad d = 1 \dots D \quad (4)$$

$$z^d = \text{MLP}(\text{LN}(\hat{z}^d)) + \hat{z}^d, \qquad d = 1 \dots D \quad (5)$$

$$p = \tanh(z_0^D W_{\text{pool}}) \quad (6)$$

ViT consists of stacked blocks that include a multiheaded self-attention (MSA) layer and an MLP layer. The position of layer normalization (LN) in ViT is the only difference from BERT: LN comes after MSA and MLP in BERT ("post-norm") and before in ViT ("pre-norm"). The input

[3] Parameters for R50 is 25M, R101 is 44M, and X152 is 60M.

[4] We also experimented with initializing the layers from BERT weights and using the pre-trained patch projection from ViT, but it did not work.

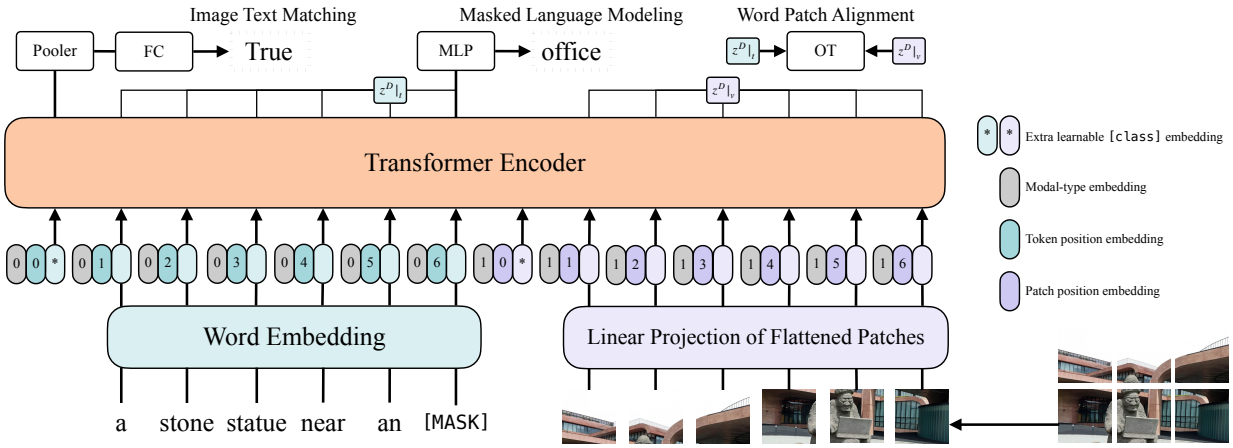



图 3. 模型概览。插图灵感来自 Dosovitskiy 等人（2020）。

推理，更不用说它还可能拖累性能。

网格特征。除了检测器头之外，ResNet 等卷积神经网络的输出特征网格也可以用作视觉-语言预训练的视觉特征。网格特征的直接使用最早由 VQA 专用模型提出（Jiang et al., 2020；Nguyen et al., 2020），主要是为了避免使用速度极慢的区域选择操作。

X-LXMERT（Cho 等，2020）通过将区域候选固定为网格，而不是使用来自区域候选网络的区域候选，重新审视了网格特征。然而，他们对特征的缓存排除了对骨干网络的进一步调优。

Pixel-BERT 是唯一一个用在 ImageNet 分类任务上预训练的 ResNet 变体骨干网络替代 VG 预训练目标检测器的 VLP 模型。与基于区域特征的 VLP 模型中冻结的检测器不同，Pixel-BERT 的骨干网络会在视觉-语言预训练期间进行调优。采用 ResNet-50 的 Pixel-BERT 在下游任务上的表现低于基于区域特征的 VLP 模型，但当使用更为庞大的 ResNeXt-152 时，其性能可与其他竞争模型相当。

然而，我们认为网格特征并非首选方案，因为深度 CNN 的计算开销仍然很大，如图 1 所示，它们占据了整体计算量的很大一部分。

补丁投影。为最大限度降低开销，我们采用最简单的视觉嵌入方案：*linear projection*，它作用于图像补丁。补丁投影嵌入由 ViT（Dosovitskiy 等，2020）为图像分类任务提出。补丁投影将视觉嵌入步骤大幅简化到文本嵌入的水平，后者同样由简单的投影（查找）操作组成。

我们使用一个 $32 \times 32$ 的 patch 投影，仅需 2.4M 参数。这与复杂的 ResNe(X)t 骨干网络[3]和检测组件形成鲜明对比。如图 1 所示，其运行时间也可以忽略不计。我们在第 4.6 节中进行了详细的运行时间分析。

## 3. 视觉-语言 Transformer

### 3.1. 模型概述

ViLT 作为一种 VLP 模型，具有简洁的架构，采用最小化的视觉嵌入流水线，并遵循单流方法。

与已有文献不同，我们使用预训练的 ViT 而非 BERT 来初始化交互 Transformer 的权重。这种初始化方式在缺少单独的深层视觉嵌入器的情况下，利用了交互层处理视觉特征的能力。[4]

$$\bar{t} = [t_{\text{class}}; t_1 T; \cdots; t_L T] + T^{\text{pos}} \tag{1}$$

$$\bar{v} = [v_{\text{class}}; v_1 V; \cdots; v_N V] + V^{\text{pos}} \tag{2}$$

$$z^0 = [\bar{t} + t^{\text{type}}; \bar{v} + v^{\text{type}}] \tag{3}$$

$$\hat{z}^d = \text{MSA}(\text{LN}(z^{d-1})) + z^{d-1}, \quad d = 1 \ldots D \tag{4}$$

$$z^d = \text{MLP}(\text{LN}(\hat{z}^d)) + \hat{z}^d, \quad d = 1 \ldots D \tag{5}$$

$$p = \tanh(z_0^D W_{\text{pool}}) \tag{6}$$

ViT 由堆叠的块组成，这些块包括一个多头自注意力（MSA）层和一个 MLP 层。ViT 中层归一化（LN）的位置是其与 BERT 的唯一区别：在 BERT 中，LN 位于 MSA 和 MLP 之后（“后归一化”），而在 ViT 中位于它们之前（“前归一化”）。输入

3R50的参数量为25M，R101为44M，X152为60M。4我们还尝试使用BERT权重初始化这些层，并使用来自ViT的预训练patch投影，但没有奏效。

text $t \in \mathbb{R}^{L\times|V|}$ is embedded to $\bar{t} \in \mathbb{R}^{L\times H}$ with a word embedding matrix $T \in \mathbb{R}^{|V|\times H}$ and a position embedding matrix $T^{\text{pos}} \in \mathbb{R}^{(L+1)\times H}$.

The input image $I \in \mathbb{R}^{C\times H\times W}$ is sliced into patches and flattened to $v \in \mathbb{R}^{N\times(P^2\cdot C)}$ where $(P, P)$ is the patch resolution and $N = HW/P^2$. Followed by linear projection $V \in \mathbb{R}^{(P^2\cdot C)\times H}$ and position embedding $V^{\text{pos}} \in \mathbb{R}^{(N+1)\times H}$, $v$ is embedded into $\bar{v} \in \mathbb{R}^{N\times H}$.

The text and image embeddings are summed with their corresponding modal-type embedding vectors $t^{\text{type}}, v^{\text{type}} \in \mathbb{R}^H$, then are concatenated into a combined sequence $z^0$. The contextualized vector $z$ is iteratively updated through $D$-depth transformer layers up until the final contextualized sequence $z^D$. $p$ is a pooled representation of the whole multimodal input, and is obtained by applying linear projection $W_{\text{pool}} \in \mathbb{R}^{H\times H}$ and hyperbolic tangent upon the first index of sequence $z^D$.

For all experiments, we use weights from ViT-B/32 pre-trained on ImageNet, hence the name ViLT-B/32.[5] Hidden size $H$ is 768, layer depth $D$ is 12, patch size $P$ is 32, MLP size is 3,072, and the number of attention heads is 12.

### 3.2. Pre-training Objectives

We train ViLT with two objectives commonly used to train VLP models: image text matching (ITM) and masked language modeling (MLM).

**Image Text Matching.** We randomly replace the aligned image with a different image with the probability of 0.5. A single linear layer ITM head projects the pooled output feature $p$ to logits over binary class, and we compute negative log-likelihood loss as our ITM loss.

Plus, inspired by the word region alignment objective in Chen et al. (2019), we design word patch alignment (WPA) that computes the alignment score between two subsets of $z^D$: $z^D|_t$ (textual subset) and $z^D|_v$ (visual subset), using the inexact proximal point method for optimal transports (IPOT) (Xie et al., 2020). We set the hyperparameters of IPOT following Chen et al. (2019) ($\beta = 0.5, N = 50$), and add the approximate wasserstein distance multiplied by 0.1 to the ITM loss.

**Masked Language Modeling.** This objective is to predict the ground truth labels of masked text tokens $t_{\text{masked}}$ from its contextualized vector $z^D_{\text{masked}}|_t$. Following the heuristics of Devlin et al. (2019), we randomly mask $t$ with the probability of 0.15.

[5] ViT-B/32 is pre-trained with ImageNet-21K and fine-tuned on ImageNet-1K for image classification. We expect that weights pre-trained on larger datasets (e.g., JFT-300M) would yield better performance.

We use a two-layer MLP MLM head that inputs $z^D_{\text{masked}}|_t$ and outputs logits over vocabulary, just as the MLM objective of BERT. The MLM loss is then computed as the negative log-likelihood loss for the masked tokens.

### 3.3. Whole Word Masking

Whole word masking is a masking technique that masks all consecutive subword tokens that compose a whole word. It is shown to be effective on downstream tasks when applied to original and Chinese BERT (Cui et al., 2019).

We hypothesize that whole word masking is particularly crucial for VLP in order to make full use of information from the other modality. For example, the word "giraffe" is tokenized into three wordpiece tokens `["gi", "##raf", "##fe"]` with the pre-trained `bert-base-uncased` tokenizer. If not all tokens are masked, say, `["gi", "[MASK]", "##fe"]`, the model may solely rely on the nearby two language tokens `["gi", "##fe"]` to predict the masked `"##raf"` rather than using the information from the image.

We mask whole words with a mask probability of 0.15 during pre-training. We discuss its impact in Section 4.5.

### 3.4. Image Augmentation

Image augmentation reportedly improves the generalization power of vision models (Shorten & Khoshgoftaar, 2019). DeiT (Touvron et al., 2020) that builds on ViT experimented with various augmentation techniques (Zhang et al., 2017; Yun et al., 2019; Berman et al., 2019; Hoffer et al., 2020; Cubuk et al., 2020), and found them beneficial for ViT training. However, the effects of image augmentation have not been explored within VLP models. Caching visual features restrains region-feature-based VLP models from using image augmentation. Notwithstanding its applicability, neither did Pixel-BERT study its effects.

To this end, we apply RandAugment (Cubuk et al., 2020) during fine-tuning. We use all the original policies except two: color inversion, because texts often contain color information as well, and cutout, as it may clear out small but important objects dispersed throughout the whole image. We use $N = 2, M = 9$ as the hyperparameters. We discuss its impact in Section 4.5 and Section 5.

## 4. Experiments

### 4.1. Overview

We use four datasets for pre-training: Microsoft COCO (MSCOCO) (Lin et al., 2014), Visual Genome (VG) (Krishna et al., 2017), SBU Captions (SBU) (Ordonez et al., 2011), and Google Conceptual Captions (GCC) (Sharma

文本 $t \in \mathbb{R}^{L\times|V|}$ 通过词嵌入矩阵 $T \in \mathbb{R}^{|V|\times H}$ 和位置嵌入矩阵 $T^{\text{pos}} \in \mathbb{R}^{(L+1)\times H}$ 嵌入到 $\bar{t} \in \mathbb{R}^{L\times H}$ 中。

输入图像 $I \in \mathbb{R}^{C\times H\times W}$ 被切分为图像块并展平成 $v \in \mathbb{R}^{N\times(P^2\cdot C)}$，其中 $(P, P)$ 是图像块分辨率，且 $N = HW/P^2$。随后经过线性投影 $V \in \mathbb{R}^{(P^2\cdot C)\times H}$ 和位置嵌入 $V^{\text{pos}} \in \mathbb{R}^{(N+1)\times H}$，$v$ 被嵌入到 $\bar{v} \in \mathbb{R}^{N\times H}$ 中。

文本和图像嵌入与其相应的模态类型嵌入向量 $t^{\text{type}}, v^{\text{type}} \in \mathbb{R}^H$ 相加，然后被拼接成一个组合序列 $z^0$。上下文化向量 $z$ 通过 $D$ 层深度的 Transformer 层迭代更新，直至得到最终的上下文化序列 $z^D$。$p$ 是整个多模态输入的池化表示，通过对序列 $z^D$ 的第一个索引应用线性投影 $W_{\text{pool}} \in \mathbb{R}^{H\times H}$ 和双曲正切函数获得。

在所有实验中，我们使用在 ImageNet 上预训练的 ViT-B/32 权重，因此命名为 ViLT-B/32。[5] 隐藏层大小 $H$ 为 768，层深度 $D$ 为 12，patch 大小 $P$ 为 32，MLP 大小为 3,072，注意力头数量为 12。

### 3.2. 预训练目标

我们使用两种常用于训练 VLP 模型的目标来训练 ViLT：图像-文本匹配（ITM）和掩码语言建模（MLM）。

图文匹配。我们以 0.5 的概率随机将对齐的图像替换为另一张不同的图像。单个线性层 ITM 头将池化输出特征 $p$ 投影为二分类的 logits，并计算负对数似然损失作为我们的 ITM 损失。

此外，受 Chen 等人（2019）中词-区域对齐目标的启发，我们设计了词-图像块对齐（WPA），使用最优传输的非精确近端点方法（IPOT）（Xie 等人，2020）来计算 $z^D$ 的两个子集之间的对齐分数：$z^D|_t$ (文本子集) 和 $z^D|_v$ (视觉子集)。我们按照 Chen 等人（2019）的设置来设定 IPOT 的超参数（$\beta = 0.5, N = 50$），并将乘以 0.1 的近似 Wasserstein 距离加到 ITM 损失中。

掩码语言建模。该目标是根据被掩码文本标记 $t_{\text{masked}}$ 的上下文化向量 $z^D_{\text{masked}}|_t$ 来预测其真实标签。遵循 Devlin 等人（2019）的启发式方法，我们以 0.15 的概率随机掩码 $t$。

5ViT-B/32使用ImageNet-21K进行预训练，并在ImageNet-1K上针对图像分类进行微调。我们预计，在更大数据集（例如JFT-300M）上预训练的权重会带来更好的性能。

我们使用一个两层 MLP 的 MLM 头，它以 $z^D_{\text{masked}}|_t$ 作为输入，并像 BERT 的 MLM 目标一样输出词表上的 logits。随后，MLM 损失被计算为被掩蔽 token 的负对数似然损失。

### 3.3. 全词掩码

全词掩码是一种掩码技术，它会掩盖构成完整词的所有连续子词标记。研究表明，将其应用于原始 BERT 和中文 BERT 时，在下游任务中是有效的（Cui 等，2019）。

我们假设，整词掩码对于 VLP 尤为关键，因为它能够充分利用来自另一模态的信息。例如，单词 “giraffe” 会被预训练的 bert-base-uncased 分词器切分为三个 wordpiece 标记 ["gi", "##raf", "##fe"]。如果并非所有标记都被掩码，例如 ["gi", "[MASK]", "##fe"]，模型可能仅依赖附近的两个语言标记 ["gi", "##fe"] 来预测被掩码的 "##raf"，而不是利用来自图像的信息。

在预训练期间，我们以 0.15 的掩码概率对整个词进行掩码。我们将在第 4.5 节讨论其影响。

### 3.4. 图像增强

据报道，图像增强能够提升视觉模型的泛化能力（Shorten & Khoshgoftaar, 2019）。基于 ViT 构建的 DeiT（Touvron et al., 2020）尝试了多种增强技术（Zhang et al., 2017; Yun et al., 2019; Berman et al., 2019; Hoffer et al., 2020; Cubuk et al., 2020），并发现它们对 ViT 训练有益。然而，图像增强的影响尚未在 VLP 模型中得到探索。缓存视觉特征限制了基于区域特征的 VLP 模型使用图像增强。尽管 Pixel-BERT 具备适用性，但它也没有研究其影响。

为此，我们在微调过程中应用 RandAugment（Cubuk 等，2020）。我们使用所有原始策略，除了两个：颜色反转，因为文本通常也包含颜色信息；以及 cutout，因为它可能会清除分散在整张图像中虽小但重要的物体。我们使用 $N = 2, M = 9$ 作为超参数。我们在第 4.5 节和第 5 节中讨论其影响。

## 4. 实验

### 4.1. 概述

我们使用四个数据集进行预训练：Microsoft COCO（MSCOCO）（Lin 等，2014）、Visual Genome（VG）（Krishna 等，2017）、SBU Captions（SBU）（Ordonez 等，2011）以及 Google Conceptual Captions（GCC）（Sharma

*Table 1.* Pre-training dataset statistics. Caption length is the length of tokens from pre-trained `bert-base-uncased` tokenizer. † GCC and SBU provide only image urls, so we collect the images from urls which were still accessible.

| Dataset | # Images | # Captions | Caption Length |
|---|---|---|---|
| MSCOCO | 113K | 567K | $11.81 \pm 2.81$ |
| VG | 108K | 5.41M | $5.53 \pm 1.76$ |
| GCC† | 3.01M | 3.01M | $10.66 \pm 4.93$ |
| SBU† | 867K | 867K | $15.0 \pm 7.74$ |

et al., 2018). Table 1 reports the dataset statistics.

We evaluate ViLT on two widely explored types of vision-and-language downstream tasks: for *classification*, we use VQAv2 (Goyal et al., 2017) and NLVR2 (Suhr et al., 2018), and for *retrieval*, we use MSCOCO and Flickr30K (F30K) (Plummer et al., 2015) re-splited by Karpathy & Fei-Fei (2015). For the classification tasks, we fine-tune three times with different initialization seeds for the head and data ordering and report the mean scores. We report the standard deviation in Table 5 along with ablation studies. For the retrieval tasks, we only fine-tune once.

### 4.2. Implementation Details

For all experiments, we use AdamW optimizer (Loshchilov & Hutter, 2018) with base learning rate of $10^{-4}$ and weight decay of $10^{-2}$. The learning rate was warmed up for 10% of the total training steps and was decayed linearly to zero for the rest of the training. Note that downstream performance may be further improved if we customize the hyperparameters to each task.

We resize the shorter edge of input images to 384 and limit the longer edge to under 640 while preserving the aspect ratio. This resizing scheme is also used during object detection in other VLP models, but with a larger size of the shorter edge (800). Patch projection of ViLT-B/32 yields 12 $\times$ 20 = 240 patches for an image with a resolution of 384 $\times$ 640. As this is a rarely reached upper limit, we sample 200 patches at maximum during pre-training. We interpolate $V^{\text{pos}}$ of ViT-B/32 to fit the size of each image and pad the patches for batch training. Note that the resulting image resolution is four times smaller than 800 $\times$ 1,333, which is the size that all other VLP models use for inputs to their visual embedders.

We use the `bert-base-uncased` tokenizer to tokenize text inputs. Instead of fine-tuning from pre-trained BERT, we learn the textual embedding-related parameters $t_{\text{class}}$, $T$, and $T^{\text{pos}}$ from scratch. Although beneficial *prima facie*, employing a pre-trained text-only BERT does not guarantee performance gain for vision and language downstream tasks. Counterevidence has already been reported by Tan & Bansal (2019), where initializing with pre-trained BERT parameters led to weaker performance than pre-training from scratch.

*Table 2.* Comparison of ViLT-B/32 with other models on downstream classification tasks. We use MCAN (Yu et al., 2019) and MaxEnt (Suhr et al., 2018) for VQAv2 and NLVR2 w/o VLP SOTA results. † additionally used GQA, VQAv2, VG-QA for pre-training. ‡ made additional use of the Open Images (Kuznetsova et al., 2020) dataset. ⓐ indicates RandAugment is applied during fine-tuning. ⊕ indicates model trained for a longer 200K pre-training steps.

| Visual Embed | Model | Time (ms) | VQAv2 test-dev | NLVR2 dev | NLVR2 test-P |
|---|---|---|---|---|---|
| Region | w/o VLP SOTA | ~900 | 70.63 | 54.80 | 53.50 |
| | ViLBERT | ~920 | 70.55 | - | - |
| | VisualBERT | ~925 | 70.80 | 67.40 | 67.00 |
| | LXMERT | ~900 | 72.42 | 74.90 | 74.50 |
| | UNITER-Base | ~900 | 72.70 | 75.85 | 75.80 |
| | OSCAR-Base† | ~900 | 73.16 | 78.07 | 78.36 |
| | VinVL-Base†‡ | ~650 | 75.95 | 82.05 | 83.08 |
| Grid | Pixel-BERT-X152 | ~160 | 74.45 | 76.50 | 77.20 |
| | Pixel-BERT-R50 | ~60 | 71.35 | 71.70 | 72.40 |
| Linear | ViLT-B/32 | ~15 | 70.33 | 74.41 | 74.57 |
| | ViLT-B/32ⓐ | ~15 | 70.85 | 74.91 | 75.57 |
| | ViLT-B/32ⓐ⊕ | ~15 | 71.26 | 75.70 | 76.13 |

We pre-train ViLT-B/32 for 100K or 200K steps on 64 NVIDIA V100 GPUs with a batch size of 4,096. For all downstream tasks, we train for ten epochs with a batch size of 256 for VQAv2/retrieval tasks and 128 for NLVR2.

### 4.3. Classification Tasks

We evaluate ViLT-B/32 on two commonly used datasets: VQAv2 and NLVR2. We use a two-layer MLP of hidden size 1,536 as the fine-tuned downstream head.

**Visual Question Answering.** The VQAv2 task asks for answers given pairs of an image and a question in natural language. The annotated answers are originally in free-form natural language, but it is a common practice to convert the task to a classification task with 3,129 answer classes. Following this practice, we fine-tune ViLT-B/32 on the VQAv2 train and validation sets while reserving 1,000 validation images and their related questions for internal validation.

We report the test-dev score results[6] from the submission to the evaluation server. ViLT falls short of VQA score compared to other VLP models with a heavy visual embedder. We suspect a detached object representation generated by the object detector eases the training of VQA since questions in VQA typically ask about objects.

[6]VQA score is calculated by comparing the inferred answer to 10 ground-truth answers: see `https://visualqa.org/evaluation.html` for details.

表 1. 预训练数据集统计。图注长度是由预训练的 bert-base-uncased 分词器得到的 token 长度。† GCC 和 SBU 仅提供图像 URL，因此我们从仍可访问的 URL 中收集图像。

| Dataset | # Images | # Captions | Caption Length |
|---|---|---|---|
| MSCOCO | 113K | 567K | 11.81 ± 2.81 |
| VG | 108K | 5.41M | 5.53 ± 1.76 |
| GCC† | 3.01M | 3.01M | 10.66 ± 4.93 |
| SBU† | 867K | 867K | 15.0 ± 7.74 |

等，2018）。表1报告了数据集统计信息。

我们在两类被广泛研究的视觉-语言下游任务上评估 ViLT：对于 *classification*，我们使用 VQAv2（Goyal 等，2017）和 NLVR2（Suhr 等，2018）；对于 *retrieval*，我们使用由 Karpathy 和 Fei-Fei（2015）重新划分的 MSCOCO 和 Flickr30K（F30K）（Plummer 等，2015）。对于分类任务，我们使用不同的头部初始化种子和数据顺序进行三次微调，并报告平均分数。我们在表 5 中报告标准差以及消融研究结果。对于检索任务，我们仅进行一次微调。

### 4.2. 实现细节

在所有实验中，我们使用 AdamW 优化器（Loshchilov & Hutter，2018），基础学习率为 $10^{-4}$，权重衰减为 $10^{-2}$。学习率在总训练步数的前 10% 进行预热，并在剩余训练过程中线性衰减至零。需要注意的是，如果我们针对每个任务定制超参数，下游性能可能会进一步提升。

我们将输入图像的较短边调整为 384，并在保持纵横比的同时将较长边限制在 640 以下。其他 VLP 模型在目标检测期间也使用这种调整大小方案，但较短边的尺寸更大（800）。对于分辨率为 384 × 640 的图像，ViLT-B/32 的补丁投影会产生 12 × 20 = 240 个补丁。由于这是一个很少达到的上限，我们在预训练期间最多采样 200 个补丁。我们对 ViT-B/32 的 $V^{\text{pos}}$ 进行插值，以适配每张图像的尺寸，并对补丁进行填充以用于批量训练。请注意，所得图像分辨率比 800 × 1,333 小四倍，而后者是所有其他 VLP 模型用于其视觉嵌入器输入的尺寸。

我们使用 bert-base-uncased 分词器对文本输入进行分词。我们没有从预训练的 BERT 进行微调，而是从头开始学习与文本嵌入相关的参数 $t_{\text{class}}$、$T$ 和 $T^{\text{pos}}$。尽管有益 *prima facie*，但采用预训练的纯文本 BERT 并不能保证提升视觉与语言下游任务的性能。Tan 和 Bansal 已经报道了反证。

表 2. ViLT-B/32 与其他模型在下游分类任务上的比较。对于不使用 VLP 的 VQAv2 和 NLVR2 SOTA 结果，我们采用 MCAN（Yu et al., 2019）和 MaxEnt（Suhr et al., 2018）。† 额外使用了 GQA、VQAv2、VG-QA 进行预训练。‡ 额外使用了 Open Images（Kuznetsova et al., 2020）数据集。a◯ 表示在微调期间应用了 RandAugment。⊕ 表示模型进行了更长的 200K 步预训练。

| Visual Embed | Model | Time (ms) | VQAv2 test-dev | NLVR2 dev | NLVR2 test-P |
|---|---|---|---|---|---|
| Region | w/o VLP SOTA | ~900 | 70.63 | 54.80 | 53.50 |
| | ViLBERT | ~920 | 70.55 | - | - |
| | VisualBERT | ~925 | 70.80 | 67.40 | 67.00 |
| | LXMERT | ~900 | 72.42 | 74.90 | 74.50 |
| | UNITER-Base | ~900 | 72.70 | 75.85 | 75.80 |
| | OSCAR-Base† | ~900 | 73.16 | 78.07 | 78.36 |
| | VinVL-Base†‡ | ~650 | 75.95 | 82.05 | 83.08 |
| Grid | Pixel-BERT-X152 | ~160 | 74.45 | 76.50 | 77.20 |
| | Pixel-BERT-R50 | ~60 | 71.35 | 71.70 | 72.40 |
| Linear | ViLT-B/32 | ~15 | 70.33 | 74.41 | 74.57 |
| | ViLT-B/32ⓐ | ~15 | 70.85 | 74.91 | 75.57 |
| | ViLT-B/32ⓐ⊕ | ~15 | 71.26 | 75.70 | 76.13 |

(2019)，其中使用预训练的 BERT 参数进行初始化导致性能弱于从头开始预训练。

我们在 64 块 NVIDIA V100 GPU 上以 4,096 的批量大小对 ViLT-B/32 进行 100K 或 200K 步的预训练。对于所有下游任务，我们训练十个 epoch，其中 VQAv2/检索任务的批量大小为 256，NLVR2 的批量大小为 128。

### 4.3. 分类任务

我们在两个常用数据集 VQAv2 和 NLVR2 上评估 ViLT-B/32。我们使用隐藏层大小为 1,536 的两层 MLP 作为微调的下游头。

视觉问答。VQAv2 任务要求根据一张图像和一个自然语言问题的配对给出答案。标注答案最初是自由形式的自然语言，但通常做法是将该任务转换为具有 3,129 个答案类别的分类任务。按照这一做法，我们在 VQAv2 的训练集和验证集上微调 ViLT-B/32，同时保留 1,000 张验证图像及其相关问题用于内部验证。

我们报告了提交到评估服务器后的 test-dev 得分结果[6]。与其他采用重型视觉嵌入器的 VLP 模型相比，ViLT 的 VQA 得分较低。我们推测，由目标检测器生成的分离对象表示有助于 VQA 的训练，因为 VQA 中的问题通常会询问有关对象的内容。

6VQA分数通过将推断出的答案与10个真实答案进行比较来计算：详情见https://visualqa.org/evaluation.html。

*Table 3.* Comparison of ViLT-B/32 with other VLP models on downstream zero-shot retrieval tasks. We exclude the models of which zero-shot retrieval performances were not reported in their original papers. † is pre-trained with a 10M proprietary vision-and-language dataset in addition to the 4M dataset of GCC+SBU. ⊕ indicates model trained for a longer 200K pre-training steps.

| Visual Embed | Model | Time (ms) | Zero-Shot Text Retrieval | | | | | | Zero-Shot Image Retrieval | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Flickr30k (1K) | | | MSCOCO (5K) | | | Flickr30k (1K) | | | MSCOCO (5K) | | |
| | | | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 |
| Region | ViLBERT | ~900 | - | - | - | - | - | - | 31.9 | 61.1 | 72.8 | - | - | - |
| | Unicoder-VL | ~925 | 64.3 | 85.8 | 92.3 | - | - | - | 48.4 | 76.0 | 85.2 | - | - | - |
| | UNITER-Base | ~900 | 80.7 | 95.7 | 98.0 | - | - | - | 66.2 | 88.4 | 92.9 | - | - | - |
| | ImageBERT† | ~925 | 70.7 | 90.2 | 94.0 | 44.0 | 71.2 | 80.4 | 54.3 | 79.6 | 87.5 | 32.3 | 59.0 | 70.2 |
| Linear | ViLT-B/32 | ~15 | 69.7 | 91.0 | 96.0 | 53.4 | 80.7 | 88.8 | 51.3 | 79.9 | 87.9 | 37.3 | 67.4 | 79.0 |
| | ViLT-B/32⊕ | ~15 | 73.2 | 93.6 | 96.5 | 56.5 | 82.6 | 89.6 | 55.0 | 82.5 | 89.8 | 40.4 | 70.0 | 81.1 |

*Table 4.* Comparison of ViLT-B/32 with other models on downstream retrieval tasks. We use SCAN for w/o VLP SOTA results. † additionally used GQA, VQAv2, VG-QA for pre-training. ‡ additionally used the Open Images dataset. ⓐ indicates RandAugment is applied during fine-tuning. ⊕ indicates model trained for a longer 200K pre-training steps.

| Visual Embed | Model | Time (ms) | Text Retrieval | | | | | | Image Retrieval | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Flickr30k (1K) | | | MSCOCO (5K) | | | Flickr30k (1K) | | | MSCOCO (5K) | | |
| | | | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 |
| Region | w/o VLP SOTA | ~900 | 67.4 | 90.3 | 95.8 | 50.4 | 82.2 | 90.0 | 48.6 | 77.7 | 85.2 | 38.6 | 69.3 | 80.4 |
| | ViLBERT-Base | ~920 | - | - | - | - | - | - | 58.2 | 84.9 | 91.5 | - | - | - |
| | Unicoder-VL | ~925 | 86.2 | 96.3 | 99.0 | 62.3 | 87.1 | 92.8 | 71.5 | 91.2 | 95.2 | 48.4 | 76.7 | 85.9 |
| | UNITER-Base | ~900 | 85.9 | 97.1 | 98.8 | 64.4 | 87.4 | 93.1 | 72.5 | 92.4 | 96.1 | 50.3 | 78.5 | 87.2 |
| | OSCAR-Base† | ~900 | - | - | - | 70.0 | 91.1 | 95.5 | - | - | - | 54.0 | 80.8 | 88.5 |
| | VinVL-Base†‡ | ~650 | - | - | - | 74.6 | 92.6 | 96.3 | - | - | - | 58.1 | 83.2 | 90.1 |
| Grid | Pixel-BERT-X152 | ~160 | 87.0 | 98.9 | 99.5 | 63.6 | 87.5 | 93.6 | 71.5 | 92.1 | 95.8 | 50.1 | 77.6 | 86.2 |
| | Pixel-BERT-R50 | ~60 | 75.7 | 94.7 | 97.1 | 59.8 | 85.5 | 91.6 | 53.4 | 80.4 | 88.5 | 41.1 | 69.7 | 80.5 |
| Linear | ViLT-B/32 | ~15 | 81.4 | 95.6 | 97.6 | 61.8 | 86.2 | 92.6 | 61.9 | 86.8 | 92.8 | 41.3 | 72.0 | 82.5 |
| | ViLT-B/32ⓐ | ~15 | 83.7 | 97.2 | 98.1 | 62.9 | 87.1 | 92.7 | 62.2 | 87.6 | 93.2 | 42.6 | 72.8 | 83.4 |
| | ViLT-B/32ⓐ⊕ | ~15 | 83.5 | 96.7 | 98.6 | 61.5 | 86.3 | 92.7 | 64.4 | 88.7 | 93.8 | 42.7 | 72.9 | 83.1 |

**Natural Language for Visual Reasoning.** The NLVR2 task is a binary classification task given triplets of two images and a question in natural language. As there are two input images unlike the pre-training setup, multiple strategies exist[7]. Following OSCAR (Li et al., 2020b) and VinVL (Zhang et al., 2021), we use the *pair* method. Here, the triplet input is reformulated into two pairs (question, image1) and (question, image2), and each pair goes through the ViLT. The head takes the concatenation of two pooled representations ($p$) as input and outputs the binary prediction.

Table 2 shows the results. ViLT-B/32 maintains competitive performance on both datasets considering its remarkable inference speed.

### 4.4. Retrieval Tasks

We fine-tune ViLT-B/32 on the Karpathy & Fei-Fei (2015) split of MSCOCO and F30K. For image-to-text and text-to-image retrieval, we measure both zero-shot and fine-tuned performance[8]. We initialize the similarity score head from the pre-trained ITM head, particularly the part that computes the true-pair logits. We sample 15 random texts as negative samples and tune the model with cross-entropy loss that maximizes the scores on positive pairs.

We report the zero shot retrieval results in Table 3 and fine-tuned results in Table 4. At zero-shot retrieval, ViLT-B/32 performs better in general than ImageBERT despite ImageBERT's pre-training on a larger (14M) dataset. At fine-tuned retrieval, recalls for ViLT-B/32 are higher by a large margin than the second fastest model (Pixel-BERT-R50).

### 4.5. Ablation Study

In Table 5, we perform various ablations. More training steps, whole word masking, and image augmentation come to be beneficial, whereas an additional training objective does not help.

It has been reported that the number of training iterations affects the performance of self-supervised models (Devlin et al., 2019; Chen et al., 2020a;b). As VLP is also a form of self-supervised training, we examine the effects of training durations. As expected, the performance constantly increases as we train the model for longer training steps (rows 1~3). Masking whole words for the MLM objective (rows 3~4) and fine-tuning with augmentation (row 6) also

[7] UNITER proposed three downstream head setups: pair, triplet, and pair-biattn.

[8] R@K corresponds to whether the ground truth is included among top K results from the validation set.

表3. ViLT-B/32 与其他 VLP 模型在下游零样本检索任务上的比较。我们排除了其原始论文中未报告零样本检索性能的模型。† 表示除 GCC+SBU 的 4M 数据集外，还使用了一个 10M 的专有视觉-语言数据集进行预训练。⊕ 表示模型经过更长的 200K 预训练步数训练。

| Visual Embed | Model | Time (ms) | Zero-Shot Text Retrieval | | | | | | Zero-Shot Image Retrieval | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Flickr30k (1K) | | | MSCOCO (5K) | | | Flickr30k (1K) | | | MSCOCO (5K) | | |
| | | | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 |
| Region | ViLBERT | ~900 | - | - | - | - | - | - | 31.9 | 61.1 | 72.8 | - | - | - |
| | Unicoder-VL | ~925 | 64.3 | 85.8 | 92.3 | - | - | - | 48.4 | 76.0 | 85.2 | - | - | - |
| | UNITER-Base | ~900 | 80.7 | 95.7 | 98.0 | - | - | - | 66.2 | 88.4 | 92.9 | - | - | - |
| | ImageBERT† | ~925 | 70.7 | 90.2 | 94.0 | 44.0 | 71.2 | 80.4 | 54.3 | 79.6 | 87.5 | 32.3 | 59.0 | 70.2 |
| Linear | ViLT-B/32 | ~15 | 69.7 | 91.0 | 96.0 | 53.4 | 80.7 | 88.8 | 51.3 | 79.9 | 87.9 | 37.3 | 67.4 | 79.0 |
| | ViLT-B/32⊕ | ~15 | 73.2 | 93.6 | 96.5 | 56.5 | 82.6 | 89.6 | 55.0 | 82.5 | 89.8 | 40.4 | 70.0 | 81.1 |

表 4. ViLT-B/32 与其他模型在下游检索任务上的比较。我们使用 SCAN 作为无 VLP 的 SOTA 结果。† 额外使用了 GQA、VQAv2、VG-QA 进行预训练。‡ 额外使用了 Open Images 数据集。a◯ 表示在微调期间应用了 RandAugment。⊕ 表示模型经过了更长的 200K 步预训练。

| Visual Embed | Model | Time (ms) | Text Retrieval | | | | | | Image Retrieval | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Flickr30k (1K) | | | MSCOCO (5K) | | | Flickr30k (1K) | | | MSCOCO (5K) | | |
| | | | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 |
| Region | w/o VLP SOTA | ~900 | 67.4 | 90.3 | 95.8 | 50.4 | 82.2 | 90.0 | 48.6 | 77.7 | 85.2 | 38.6 | 69.3 | 80.4 |
| | ViLBERT-Base | ~920 | - | - | - | - | - | - | 58.2 | 84.9 | 91.5 | - | - | - |
| | Unicoder-VL | ~925 | 86.2 | 96.3 | 99.0 | 62.3 | 87.1 | 92.8 | 71.5 | 91.2 | 95.2 | 48.4 | 76.7 | 85.9 |
| | UNITER-Base | ~900 | 85.9 | 97.1 | 98.8 | 64.4 | 87.4 | 93.1 | 72.5 | 92.4 | 96.1 | 50.3 | 78.5 | 87.2 |
| | OSCAR-Base† | ~900 | - | - | - | 70.0 | 91.1 | 95.5 | - | - | - | 54.0 | 80.8 | 88.5 |
| | VinVL-Base†‡ | ~650 | - | - | - | 74.6 | 92.6 | 96.3 | - | - | - | 58.1 | 83.2 | 90.1 |
| Grid | Pixel-BERT-X152 | ~160 | 87.0 | 98.9 | 99.5 | 63.6 | 87.5 | 93.6 | 71.5 | 92.1 | 95.8 | 50.1 | 77.6 | 86.2 |
| | Pixel-BERT-R50 | ~60 | 75.7 | 94.7 | 97.1 | 59.8 | 85.5 | 91.6 | 53.4 | 80.4 | 88.5 | 41.1 | 69.7 | 80.5 |
| Linear | ViLT-B/32 | ~15 | 81.4 | 95.6 | 97.6 | 61.8 | 86.2 | 92.6 | 61.9 | 86.8 | 92.8 | 41.3 | 72.0 | 82.5 |
| | ViLT-B/32ⓐ | ~15 | 83.7 | 97.2 | 98.1 | 62.9 | 87.1 | 92.7 | 62.2 | 87.6 | 93.2 | 42.6 | 72.8 | 83.4 |
| | ViLT-B/32ⓐ⊕ | ~15 | 83.5 | 96.7 | 98.6 | 61.5 | 86.3 | 92.7 | 64.4 | 88.7 | 93.8 | 42.7 | 72.9 | 83.1 |

用于视觉推理的自然语言。NLVR2 任务是一个二分类任务，输入为由两张图像和一个自然语言问题组成的三元组。由于与预训练设置不同，该任务有两张输入图像，因此存在多种策略[7]。遵循 OSCAR（Li et al., 2020b）和 VinVL（Zhang et al., 2021），我们使用 *pair* 方法。在这里，三元组输入被重新表述为两个配对：（问题，图像1）和（问题，图像2），每个配对都经过 ViLT。分类头将两个池化表示（$p$）的拼接作为输入，并输出二分类预测。

表 2 展示了结果。考虑到其卓越的推理速度，ViLT-B/32 在两个数据集上都保持了具有竞争力的性能。

### 4.4. 检索任务

我们在 MSCOCO 和 F30K 的 Karpathy & Fei-Fei（2015）划分上对 ViLT-B/32 进行微调。对于图像到文本和文本到图像检索，我们同时衡量零样本和微调后的性能[8]。我们从以下来源初始化相似度评分头

7UNITER提出了三种下游头设置：pair、triplet和pair-biattn。8R@K对应于真实结果是否包含在验证集的前K个结果中。

预训练的 ITM 头，尤其是计算真实配对 logits 的部分。我们随机采样 15 个文本作为负样本，并使用交叉熵损失对模型进行调优，以最大化正样本对的分数。

我们在表3中报告了零样本检索结果，在表4中报告了微调结果。在零样本检索中，尽管 ImageBERT 在更大的（14M）数据集上进行了预训练，但 ViLT-B/32 的整体表现优于 ImageBERT。在微调检索中，ViLT-B/32 的召回率大幅高于速度第二快的模型（Pixel-BERT-R50）。

### 4.5. 消融研究

在表 5 中，我们进行了各种消融实验。更多的训练步数、全词掩码和图像增强被证明是有益的，而额外的训练目标则没有帮助。

据报道，训练迭代次数会影响自监督模型的性能（Devlin 等，2019；Chen 等，2020a；b）。由于 VLP 也是一种自监督训练形式，我们考察了训练时长的影响。正如预期的那样，随着我们用更长的训练步数训练模型，性能持续提升（第 1~3 行）。针对 MLM 目标进行整词掩码（第 3~4 行）以及使用增强进行微调（第 6 行）也

*Table 5.* Ablation study of ViLT-B/32. ⓦ denotes whether whole word masking is used for pre-training. ⓜ denotes whether MPP objective is used for pre-training. ⓐ denotes whether RandAugment is used during fine-tuning.

| Training Steps | Ablation ⓦ | ⓜ | ⓐ | VQAv2 test-dev | NLVR2 dev | NLVR2 test-P | Flickr30k R@1 (1K) TR (ZS) | Flickr30k R@1 (1K) IR (ZS) | MSCOCO R@1 (5K) TR (ZS) | MSCOCO R@1 (5K) IR (ZS) |
|---|---|---|---|---|---|---|---|---|---|---|
| 25K | X | X | X | 68.96 ± 0.07 | 70.83 ± 0.19 | 70.83 ± 0.23 | 75.39 (45.12) | 52.52 (31.80) | 53.72 (31.55) | 34.88 (21.58) |
| 50K | X | X | X | 69.80 ± 0.01 | 71.93 ± 0.27 | 72.92 ± 0.82 | 78.13 (55.57) | 57.36 (40.94) | 57.00 (39.56) | 37.47 (27.51) |
| 100K | X | X | X | 70.16 ± 0.01 | 73.54 ± 0.02 | 74.15 ± 0.27 | 79.39 (66.99) | 60.50 (47.62) | 60.15 (51.25) | 40.45 (34.59) |
| 100K | O | X | X | 70.33 ± 0.01 | 74.41 ± 0.21 | 74.57 ± 0.09 | 81.35 (69.73) | 61.86 (51.28) | 61.79 (53.40) | 41.25 (37.26) |
| 100K | O | O | X | 70.21 ± 0.05 | 72.76 ± 0.50 | 73.54 ± 0.47 | 78.91 (63.67) | 58.76 (46.96) | 59.53 (47.75) | 40.08 (32.28) |
| 100K | O | X | O | 70.85 ± 0.13 | 74.91 ± 0.29 | 75.57 ± 0.61 | 83.69 (69.73) | 62.22 (51.28) | 62.88 (53.40) | 42.62 (37.26) |
| 200K | O | X | O | 71.26 ± 0.06 | 75.70 ± 0.32 | 76.13 ± 0.39 | 83.50 (73.24) | 64.36 (54.96) | 61.49 (56.51) | 42.70 (40.42) |

*Table 6.* Comparison of VLP models in terms of parameter size, FLOPs, and inference latency. Since FLOPs are proportional to input size, we denote the number of input tokens (image+text) as superscripts ("?" when text length is unreported; we arbitrarily use length 40). Although not captured in FLOPs count nor parameter size (because it is not a tensor operation), note that *per-class* NMS for 1,600 classes amounts to more than 500 ms in latency. NMS latency varies a lot according to the number of detected classes.

| Visual Embed | Model | #Params (M) | #FLOPs (G) | Time (ms) |
|---|---|---|---|---|
| Region | ViLBERT$^{36+36}$ | 274.3 | 958.1 | ~900 |
| | VisualBERT$^{36+128}$ | 170.3 | 425.0 | ~925 |
| | LXMERT$^{36+20}$ | 239.8 | 952.0 | ~900 |
| | UNITER-Base$^{36+60}$ | 154.7 | 949.9 | ~900 |
| | OSCAR-Base$^{50+35}$ | 154.7 | 956.4 | ~900 |
| | VinVL-Base$^{50+35}$ | 157.3 | 1023.3 | ~650 |
| | Unicoder-VL$^{100+?}$ | 170.3 | 419.7 | ~925 |
| | ImageBERT$^{100+44}$ | 170.3 | 420.6 | ~925 |
| Grid | Pixel-BERT-X152$^{146+?}$ | 144.3 | 185.8 | ~160 |
| | Pixel-BERT-R50$^{260+?}$ | 94.9 | 136.8 | ~60 |
| Linear | ViLT-B/32$^{200+40}$ | 87.4 | 55.9 | ~15 |

drive performance. Further increase in training iterations to 200K improved performance on VQAv2, NLVR2, and zero-shot retrieval. We stop increasing the number of iterations over 200K as the fine-tuned text retrieval performance decreases afterward.

An additional masked region modeling (MRM) objective has been the key for performance boost in VLP models such as Chen et al. (2019). We experiment with masked patch prediction (MPP) (Dosovitskiy et al., 2020) which mimics the effect of MRM in a form compatible with patch projections. The patch $v$ is masked with the probability of 0.15, and the model predicts the mean RGB value of the masked patch from its contextualized vector $z^D_{\text{masked}|v}$. However, MPP turns out not to be contributing to downstream performance (rows 4~5). This result is in sharp contrast to the MRM objective on supervision signals from object detection.

*Table 7.* VLP model components. "PC" is for *per-class* manner NMS and "CA" is for *class-agnostic*. Following Tan & Bansal (2019), one single-modality layer is counted as 0.5 multi-modality layer.

| Visual Embed | Model | CNN Backbone | RoI Head | NMS | Trans. Layers |
|---|---|---|---|---|---|
| Region | ViLBERT | R101 | C4 | PC | ~15 |
| | VisualBERT | X152 | FPN | PC | 12 |
| | LXMERT | R101 | C4 | PC | ~12 |
| | UNITER-Base | R101 | C4 | PC | 12 |
| | OSCAR-Base | R101 | C4 | PC | 12 |
| | VinVL-Base | X152 | C4 | CA | 12 |
| | Unicoder-VL | X152 | FPN | PC | 12 |
| | ImageBERT | X152 | FPN | PC | 12 |
| Grid | Pixel-BERT-X152 | X152 | - | - | 12 |
| | Pixel-BERT-R50 | R50 | - | - | 12 |
| Linear | ViLT-B/32 | - | - | - | 12 |

### 4.6. Complexity Analysis of VLP Models

We analyze the complexity of VLP models in various terms. In Table 6, we report the number of parameters, the number of floating-point operations (FLOPs), and the inference latency of the visual embedder and transformer. We exclude the textual embedder because it is shared by all VLP models[9]. The latency is averaged over 10K times on a Xeon E5-2650 CPU and an NVIDIA P40 GPU.

The input size in terms of image resolution and the length of concatenated multimodal input sequence affects the number of FLOPs. We co-note the sequence lengths. The image resolution is 800 × 1,333 for region-based VLP models and Pixel-BERT-R50, 600 × 1,000 for Pixel-BERT-X152, and 384 × 640 for ViLT-B/32.

In Pixel-BERT and ViLT, visual tokens are sampled during pre-training and used in full during fine-tuning. We report the maximum number of visual tokens.

We observe that the runtime of BERT-base-like transformers varies only by $< 1$ ms for input sequences of length under 300. Since patch projection of ViLT-B/32 generates at most

[9]FLOPs and time are neglectable because the operation is an embedding lookup. The 30K embedding dictionary used by `bert-base-uncased` has 23.47 M parameters

表5. ViLT-B/32 的消融研究。w◯ 表示预训练时是否使用全词掩码。m◯ 表示预训练时是否使用 MPP 目标。a◯ 表示微调期间是否使用 RandAugment。

| Training Steps | Ablation ⓦ | ⓜ | ⓐ | VQAv2 test-dev | NLVR2 dev | test-P | Flickr30k R@1 (1K) TR (ZS) | IR (ZS) | MSCOCO R@1 (5K) TR (ZS) | IR (ZS) |
|---|---|---|---|---|---|---|---|---|---|---|
| 25K | X | X | X | 68.96 ± 0.07 | 70.83 ± 0.19 | 70.83 ± 0.23 | 75.39 (45.12) | 52.52 (31.80) | 53.72 (31.55) | 34.88 (21.58) |
| 50K | X | X | X | 69.80 ± 0.01 | 71.93 ± 0.27 | 72.92 ± 0.82 | 78.13 (55.57) | 57.36 (40.94) | 57.00 (39.56) | 37.47 (27.51) |
| 100K | X | X | X | 70.16 ± 0.01 | 73.54 ± 0.02 | 74.15 ± 0.27 | 79.39 (66.99) | 60.50 (47.62) | 60.15 (51.25) | 40.45 (34.59) |
| 100K | O | X | X | 70.33 ± 0.01 | 74.41 ± 0.21 | 74.57 ± 0.09 | 81.35 (69.73) | 61.86 (51.28) | 61.79 (53.40) | 41.25 (37.26) |
| 100K | O | O | X | 70.21 ± 0.05 | 72.76 ± 0.50 | 73.54 ± 0.47 | 78.91 (63.67) | 58.76 (46.96) | 59.53 (47.75) | 40.08 (32.28) |
| 100K | O | X | O | 70.85 ± 0.13 | 74.91 ± 0.29 | 75.57 ± 0.61 | 83.69 (69.73) | 62.22 (51.28) | 62.88 (53.40) | 42.62 (37.26) |
| 200K | O | X | O | 71.26 ± 0.06 | 75.70 ± 0.32 | 76.13 ± 0.39 | 83.50 (73.24) | 64.36 (54.96) | 61.49 (56.51) | 42.70 (40.42) |

表 6. 从参数规模、FLOPs 和推理延迟方面比较 VLP 模型。由于 FLOPs 与输入大小成正比，我们将输入 token 数量（图像+文本）表示为上标（当文本长度未报告时用“?”表示；我们任意使用长度 40）。尽管 FLOPs 计数和参数规模未涵盖这一点（因为它不是张量运算），但请注意，*per-class* 对 1,600 个类别执行 NMS 会带来超过 500 ms 的延迟。NMS 延迟会根据检测到的类别数量而有很大差异。

| Visual Embed | Model | #Params (M) | #FLOPs (G) | Time (ms) |
|---|---|---|---|---|
| Region | ViLBERT$^{36+36}$ | 274.3 | 958.1 | ~900 |
| | VisualBERT$^{36+128}$ | 170.3 | 425.0 | ~925 |
| | LXMERT$^{36+20}$ | 239.8 | 952.0 | ~900 |
| | UNITER-Base$^{36+60}$ | 154.7 | 949.9 | ~900 |
| | OSCAR-Base$^{50+35}$ | 154.7 | 956.4 | ~900 |
| | VinVL-Base$^{50+35}$ | 157.3 | 1023.3 | ~650 |
| | Unicoder-VL$^{100+?}$ | 170.3 | 419.7 | ~925 |
| | ImageBERT$^{100+44}$ | 170.3 | 420.6 | ~925 |
| Grid | Pixel-BERT-X152$^{146+?}$ | 144.3 | 185.8 | ~160 |
| | Pixel-BERT-R50$^{260+?}$ | 94.9 | 136.8 | ~60 |
| Linear | ViLT-B/32$^{200+40}$ | 87.4 | 55.9 | ~15 |

驱动性能。将训练迭代次数进一步增加到 200K 后，在 VQAv2、NLVR2 和零样本检索上的性能得到提升。由于之后微调后的文本检索性能会下降，我们停止将迭代次数增加到 200K 以上。

额外的掩码区域建模（MRM）目标一直是提升 VLP 模型性能的关键，例如 Chen 等人（2019）。我们尝试了掩码图块预测（MPP）（Dosovitskiy 等人，2020），它以一种与图块投影兼容的形式模拟 MRM 的效果。图块 $v$ 以 0.15 的概率被掩码，模型根据其上下文化向量 $z^D_{\text{masked}}|_v$ 预测被掩码图块的平均 RGB 值。然而，事实证明 MPP 并未对下游性能有所贡献（第 4~5 行）。这一结果与基于目标检测监督信号的 MRM 目标形成了鲜明对比。

表 7. VLP 模型组件。“PC”表示 *per-class* 方式的 NMS，“CA”表示 *class-agnostic*。按照 Tan & Bansal (2019)，一个单模态层计为 0.5 个多模态层。

| Visual Embed | Model | CNN Backbone | RoI Head | NMS | Trans. Layers |
|---|---|---|---|---|---|
| Region | ViLBERT | R101 | C4 | PC | ~15 |
| | VisualBERT | X152 | FPN | PC | 12 |
| | LXMERT | R101 | C4 | PC | ~12 |
| | UNITER-Base | R101 | C4 | PC | 12 |
| | OSCAR-Base | R101 | C4 | PC | 12 |
| | VinVL-Base | X152 | C4 | CA | 12 |
| | Unicoder-VL | X152 | FPN | PC | 12 |
| | ImageBERT | X152 | FPN | PC | 12 |
| Grid | Pixel-BERT-X152 | X152 | - | - | 12 |
| | Pixel-BERT-R50 | R50 | - | - | 12 |
| Linear | ViLT-B/32 | - | - | - | 12 |

### 4.6. VLP 模型的复杂度分析

我们从多个方面分析 VLP 模型的复杂度。在表 6 中，我们报告了视觉嵌入器和 Transformer 的参数数量、浮点运算次数（FLOPs）以及推理延迟。我们排除了文本嵌入器，因为它被所有 VLP 模型共享[9]。延迟是在 Xeon E5-2650 CPU 和 NVIDIA P40 GPU 上进行 1 万次测试后取平均得到的。

以图像分辨率表示的输入大小以及拼接后的多模态输入序列长度会影响 FLOPs 数量。我们同时标注序列长度。对于基于区域的 VLP 模型和 Pixel-BERT-R50，图像分辨率为 800 × 1,333；对于 Pixel-BERT-X152，为 600 × 1,000；对于 ViLT-B/32，为 384 × 640。

在 Pixel-BERT 和 ViLT 中，视觉 token 在预训练期间进行采样，并在微调期间全部使用。我们报告视觉 token 的最大数量。

我们观察到，对于长度小于 300 的输入序列，类似 BERT-base 的 Transformer 的运行时间变化仅为 $< 1$ 毫秒。由于 ViLT-B/32 的图像块投影最多生成

9FLOPs和时间可以忽略不计，因为该操作是一次嵌入查找。bert-base-uncased使用的30K嵌入字典有23.47 M个参数。

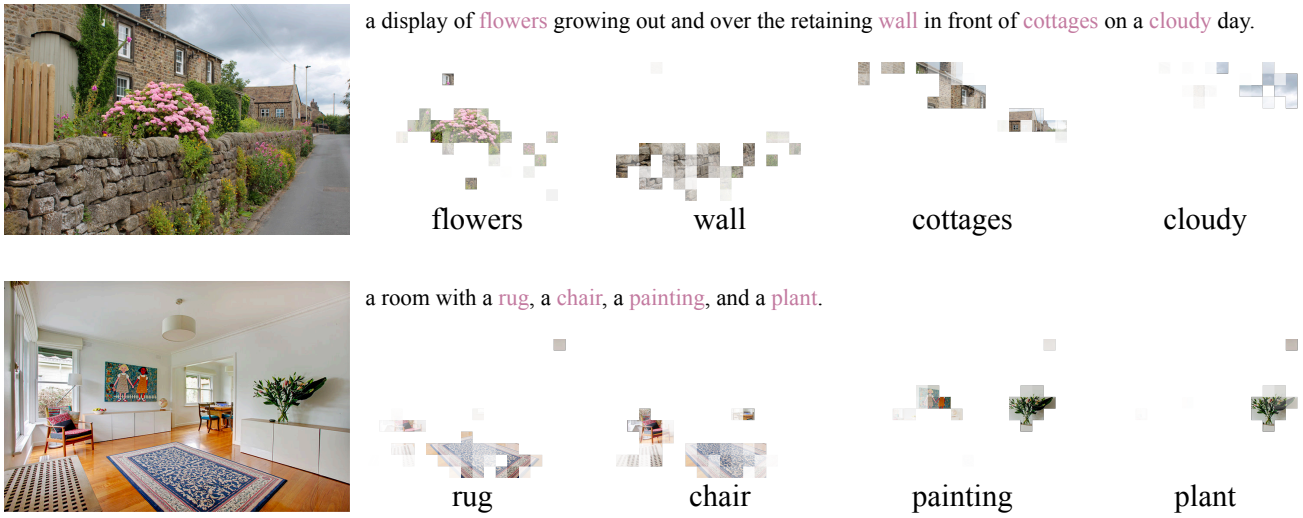



*Figure 4.* Visualizations of transportation plan of word patch alignment. Best viewed zoomed in.

240 image tokens, our model can still be efficient even though it receives a combination of image and text tokens.

### 4.7. Visualization

Figure 4 is an example of a cross-modal alignment. The transportation plan of WPA expresses a heatmap for a text token highlighted in pink color. Each square tile represents a patch, and its opacity indicates how much mass is transported from the highlighted word token.

More IPOT iterations– more than over 50 as in the training phase– help the visualization heatmap converge; empirically, 1,000 iterations are sufficient to get a clearly identifiable heatmap. We z-normalize the plan for each token and clamp the values to [1.0, 3.0].

## 5. Conclusion and Future Work

In this paper, we present a minimal VLP architecture, Vision-and-Langauge Transformer (ViLT). ViLT is competent to competitors which are heavily equipped with convolutional visual embedding networks (e.g., Faster R-CNN and ResNets). We ask for future work on VLP to focus more on the modality interactions inside the transformer module rather than engaging in an arms race that merely powers up unimodal embedders.

Although remarkable as it is, ViLT-B/32 is more of a proof of concept that efficient VLP models free of convolution and region supervision can still be competent. We wrap up by pointing out a few factors that may add to the ViLT family.

**Scalability.** As shown in papers on large-scale transformers (Devlin et al., 2019; Dosovitskiy et al., 2020), the performance of pre-trained transformers scale well given an appropriate amount of data. This observation paves the way for even better performing ViLT variants (e.g., ViLT-L (large) and ViLT-H (huge)). We leave training larger models for future work because aligned vision-and-language datasets are yet scarce.

**Masked Modeling for Visual Inputs.** Considering the success of MRM, we speculate that the masked modeling objective for the visual modality helps by preserving the information up until the last layer of the transformer. However, as observed in Table 5, a naive variant of MRM on image patches (MPP) fails.

Cho et al. (2020) proposed to train their grid RoIs on masked object classification (MOC) tasks. However, the visual vocabulary cluster in this work was fixed during the vision and language pre-training together with the visual backbone. For trainable visual embedders, one-time clustering is not a viable option. We believe that alternating clustering (Caron et al., 2018; 2019) or simultaneous clustering (Asano et al., 2019; Caron et al., 2020) methods studied in visual unsupervised learning research could be applied.

We encourage future work that does not use region supervision to devise a more sophisticated masking objective for the visual modality.

**Augmentation Strategies.** Previous work on contrastive visual representation learning (Chen et al., 2020a;b) showed that gaussian blur, not employed by RandAugment, brings noticeable gains to downstream performance compared with a simpler augmentation strategy (He et al., 2020). Exploration of appropriate augmentation strategies for textual and visual inputs would be a valuable addition.

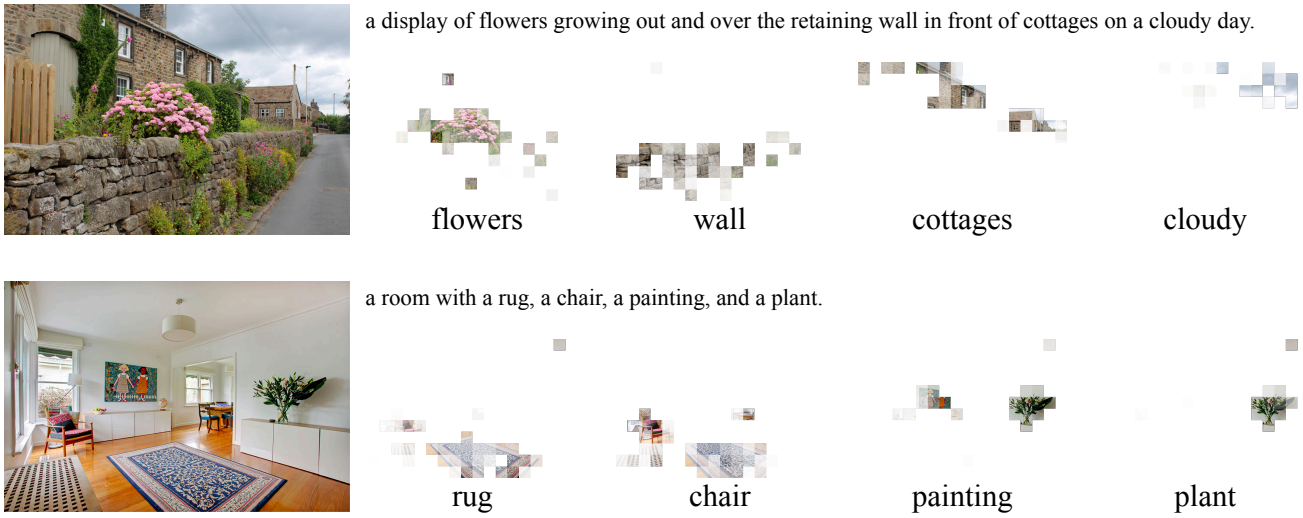



图 4. 词块对齐的传输计划可视化。放大查看效果最佳。

240 个图像 token，即便接收的是图像 token 与文本 token 的组合，我们的模型仍能保持高效。

### 4.7. 可视化

图 4 是跨模态对齐的一个示例。WPA 的传输计划展示了以粉色高亮的文本词元的热力图。每个方形小块代表一个补丁，其不透明度表示从高亮词元传输的质量大小。

更多的 IPOT 迭代次数——超过训练阶段的 50 次以上——有助于可视化热图收敛；根据经验，1,000 次迭代足以得到清晰可辨的热图。我们对每个 token 的方案进行 z 归一化，并将数值截断到 [1.0, 3.0]。

## 5. 结论与未来工作

在本文中，我们提出了一种极简的 VLP 架构——视觉与语言 Transformer（Vision-and-Langauge Transformer，ViLT）。ViLT 足以与那些大量配备卷积视觉嵌入网络（例如 Faster R-CNN 和 ResNets）的竞争模型相媲美。我们呼吁未来关于 VLP 的研究更多关注 Transformer 模块内部的模态交互，而不是陷入仅仅强化单模态嵌入器的军备竞赛。

尽管 ViLT-B/32 已经相当出色，但它更像是一个概念验证，表明不依赖卷积和区域监督的高效 VLP 模型仍然可以具备竞争力。最后，我们指出一些可能为 ViLT 系列带来改进的因素。

可扩展性。正如关于大规模 Transformer 的论文所示（Devlin 等，2019；Dosovitskiy 等，2020），每个-在给定适量数据的情况下，预训练 Transformer 的性能能够很好地扩展。这一观察为性能更优的 ViLT 变体（例如 ViLT-L（large）和 ViLT-H（huge））铺平了道路。由于对齐的视觉-语言数据集仍然稀缺，我们将训练更大的模型留作未来工作。

视觉输入的掩码建模。鉴于 MRM 的成功，我们推测，视觉模态的掩码建模目标通过将信息保留到 Transformer 的最后一层而发挥作用。然而，如表 5 所示，在图像块上直接采用 MRM 的朴素变体（MPP）并不奏效。

Cho 等人（2020）提出在掩码目标分类（MOC）任务上训练其网格 RoI。然而，在该工作中，视觉词汇聚类在视觉与语言预训练期间与视觉骨干网络一起被固定。对于可训练的视觉嵌入器而言，一次性聚类并不是一个可行的选择。我们认为，视觉无监督学习研究中所研究的交替聚类（Caron 等，2018；2019）或同步聚类（Asano 等，2019；Caron 等，2020）方法可以被应用。

我们鼓励未来不使用区域监督的工作为视觉模态设计更复杂的掩码目标。

增强策略。先前关于对比式视觉表征学习的工作（Chen et al., 2020a;b）表明，与更简单的增强策略相比（He et al., 2020），RandAugment 未采用的高斯模糊能为下游性能带来显著提升。探索适用于文本和视觉输入的合适增强策略将是一个有价值的补充。

## References


Anderson, P., He, X., Buehler, C., Teney, D., Johnson, M., Gould, S., and Zhang, L. Bottom-up and top-down attention for image captioning and visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6077–6086, 2018.

Asano, Y., Rupprecht, C., and Vedaldi, A. Self-labelling via simultaneous clustering and representation learning. In *International Conference on Learning Representations*, 2019.

Berman, M., Jégou, H., Vedaldi, A., Kokkinos, I., and Douze, M. Multigrain: a unified image embedding for classes and instances. *arXiv preprint arXiv:1902.05509*, 2019.

Bugliarello, E., Cotterell, R., Okazaki, N., and Elliott, D. Multimodal pretraining unmasked: Unifying the vision and language berts. *arXiv preprint arXiv:2011.15124*, 2020.

Caron, M., Bojanowski, P., Joulin, A., and Douze, M. Deep clustering for unsupervised learning of visual features. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 132–149, 2018.

Caron, M., Bojanowski, P., Mairal, J., and Joulin, A. Unsupervised pre-training of image features on non-curated data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 2959–2968, 2019.

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., and Joulin, A. Unsupervised learning of visual features by contrasting cluster assignments. *arXiv preprint arXiv:2006.09882*, 2020.

Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pp. 1597–1607. PMLR, 2020a.

Chen, X., Fan, H., Girshick, R., and He, K. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020b.

Chen, Y.-C., Li, L., Yu, L., Kholy, A. E., Ahmed, F., Gan, Z., Cheng, Y., and Liu, J. Uniter: Learning universal image-text representations. *arXiv preprint arXiv:1909.11740*, 2019.

Cho, J., Lu, J., Schwenk, D., Hajishirzi, H., and Kembhavi, A. X-lxmert: Paint, caption and answer questions with multi-modal transformers. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 8785–8805, 2020.

Cubuk, E. D., Zoph, B., Shlens, J., and Le, Q. V. Randaugment: Practical automated data augmentation with a reduced search space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pp. 702–703, 2020.

Cui, Y., Che, W., Liu, T., Qin, B., Yang, Z., Wang, S., and Hu, G. Pre-training with whole word masking for chinese bert. *arXiv preprint arXiv:1906.08101*, 2019.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4171–4186, 2019.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

Faghri, F., Fleet, D. J., Kiros, J. R., and Fidler, S. Vse++: Improving visual-semantic embeddings with hard negatives. *arXiv preprint arXiv:1707.05612*, 2017.

Gan, Z., Chen, Y.-C., Li, L., Zhu, C., Cheng, Y., and Liu, J. Large-scale adversarial training for vision-and-language representation learning. *arXiv preprint arXiv:2006.06195*, 2020.

Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., and Parikh, D. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 6904–6913, 2017.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.

He, K., Gkioxari, G., Dollár, P., and Girshick, R. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pp. 2961–2969, 2017.

He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9729–9738, 2020.

Hoffer, E., Ben-Nun, T., Hubara, I., Giladi, N., Hoefler, T., and Soudry, D. Augment your batch: Improving generalization through instance repetition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8129–8138, 2020.

## 参考文献


Anderson, P., He, X., Buehler, C., Teney, D., Johnson, M., Gould, S., 和 Zhang, L. 用于图像描述生成和视觉问答的自底向上与自顶向下注意力。在 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 6077–6086 页，2018。

Asano, Y., Rupprecht, C., 和 Vedaldi, A. 通过同时聚类和表示学习进行自标注。在 *International Conference on Learning Representations*，2019。

Berman, M., Jégou, H., Vedaldi, A., Kokkinos, I., 和 Douze, M. Multigrain：用于类别和实例的统一图像嵌入。*arXiv preprint arXiv:1902.05509*，2019。

Bugliarello, E., Cotterell, R., Okazaki, N., 和 Elliott, D. 揭开多模态预训练的面纱：统一视觉与语言 BERT。*arXiv preprint arXiv:2011.15124*，2020。

Caron, M., Bojanowski, P., Joulin, A., 和 Douze, M. 用于视觉特征无监督学习的深度聚类。在 *Proceedings of the European Conference on Computer Vision (ECCV)*，第 132–149 页，2018。

Caron, M., Bojanowski, P., Mairal, J., 和 Joulin, A. 在非精选数据上进行图像特征的无监督预训练。在 *Proceedings of the IEEE/CVF International Conference on Computer Vision*，第 2959–2968 页，2019。

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., 和 Joulin, A. 通过对比聚类分配进行视觉特征的无监督学习。*arXiv preprint arXiv:2006.09882*，2020。

Chen, T., Kornblith, S., Norouzi, M., 和 Hinton, G. 用于视觉表示对比学习的简单框架。在 *International conference on machine learning*，第 1597–1607 页。PMLR，2020a。

Chen, X., Fan, H., Girshick, R., 和 He, K. 使用动量对比学习改进基线。*arXiv preprint arXiv:2003.04297*，2020b。

Chen, Y.-C., Li, L., Yu, L., Kholy, A. E., Ahmed, F., Gan, Z., Cheng, Y., 和 Liu, J. Uniter：学习通用图像-文本表示。*arXiv preprint arXiv:1909.11740*，2019。

Cho, J.、Lu, J.、Schwenk, D.、Hajishirzi, H. 和 Kembhavi, A. X-lxmert：使用多模态 Transformer 进行绘画、描述并回答问题。载于 *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*，第 8785–8805 页，2020。

Cubuk, E. D., Zoph, B., Shlens, J., 和 Le, Q. V. Randaugment：具有缩减搜索空间的实用自动化数据增强。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*，第 702–703 页，2020。

Cui, Y., Che, W., Liu, T., Qin, B., Yang, Z., Wang, S., and Hu, G. 用于中文 BERT 的全词掩码预训练。*arXiv preprint arXiv:1906.08101*, 2019.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. Bert：用于语言理解的深度双向 Transformer 预训练。见 *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*，第 4171–4186 页，2019。

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., 等。一张图像价值 16x16 个词：用于大规模图像识别的 Transformer。*arXiv preprint arXiv:2010.11929*, 2020.

Faghri, F., Fleet, D. J., Kiros, J. R., 和 Fidler, S. Vse++：利用困难负样本改进视觉-语义嵌入。*arXiv preprint arXiv:1707.05612*, 2017.

Gan, Z., Chen, Y.-C., Li, L., Zhu, C., Cheng, Y., 和 Liu, J. 用于视觉与语言表示学习的大规模对抗训练。*arXiv preprint arXiv:2006.06195*, 2020.

Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., 和 Parikh, D. 让 VQA 中的 V 发挥作用：提升图像理解在视觉问答中的作用。见 *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*，第 6904–6913 页，2017 年。

He, K., Zhang, X., Ren, S., 和 Sun, J. 用于图像识别的深度残差学习。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 770–778 页，2016。

He, K.、Gkioxari, G.、Dollár, P. 和 Girshick, R. Mask R-CNN。载于 *Proceedings of the IEEE international conference on computer vision*，第 2961–2969 页，2017 年。

He, K.、Fan, H.、Wu, Y.、Xie, S. 和 Girshick, R. 用于无监督视觉表示学习的动量对比。在 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 9729–9738 页，2020 年。

Hoffer, E., Ben-Nun, T., Hubara, I., Giladi, N., Hoefler, T., 和 Soudry, D. 扩增你的批次：通过实例重复提高泛化能力。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 8129–8138 页，2020 年。

Huang, Z., Zeng, Z., Liu, B., Fu, D., and Fu, J. Pixel-bert: Aligning image pixels with text by deep multi-modal transformers. *arXiv preprint arXiv:2004.00849*, 2020.

Jiang, H., Misra, I., Rohrbach, M., Learned-Miller, E., and Chen, X. In defense of grid features for visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10267–10276, 2020.

Jiang, Y., Natarajan, V., Chen, X., Rohrbach, M., Batra, D., and Parikh, D. Pythia v0. 1: the winning entry to the vqa challenge 2018. *arXiv preprint arXiv:1807.09956*, 2018.

Karpathy, A. and Fei-Fei, L. Deep visual-semantic alignments for generating image descriptions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3128–3137, 2015.

Krishna, R., Zhu, Y., Groth, O., Johnson, J., Hata, K., Kravitz, J., Chen, S., Kalantidis, Y., Li, L.-J., Shamma, D. A., et al. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International journal of computer vision*, 123(1):32–73, 2017.

Krizhevsky, A., Sutskever, I., and Hinton, G. E. Imagenet classification with deep convolutional neural networks. In *NIPS*, 2012.

Kuznetsova, A., Rom, H., Alldrin, N., Uijlings, J., Krasin, I., Pont-Tuset, J., Kamali, S., Popov, S., Malloci, M., Kolesnikov, A., et al. The open images dataset v4. *International Journal of Computer Vision*, pp. 1–26, 2020.

Lee, K.-H., Chen, X., Hua, G., Hu, H., and He, X. Stacked cross attention for image-text matching. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 201–216, 2018.

Li, G., Duan, N., Fang, Y., Gong, M., Jiang, D., and Zhou, M. Unicoder-vl: A universal encoder for vision and language by cross-modal pre-training. In *AAAI*, pp. 11336–11344, 2020a.

Li, L. H., Yatskar, M., Yin, D., Hsieh, C.-J., and Chang, K.-W. Visualbert: A simple and performant baseline for vision and language. *arXiv preprint arXiv:1908.03557*, 2019.

Li, X., Yin, X., Li, C., Zhang, P., Hu, X., Zhang, L., Wang, L., Hu, H., Dong, L., Wei, F., et al. Oscar: Object-semantics aligned pre-training for vision-language tasks. In *European Conference on Computer Vision*, pp. 121–137. Springer, 2020b.

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., and Zitnick, C. L. Microsoft coco: Common objects in context. In *European conference on computer vision*, pp. 740–755. Springer, 2014.

Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2018.

Lu, J., Batra, D., Parikh, D., and Lee, S. Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. In *Advances in Neural Information Processing Systems*, pp. 13–23, 2019.

Lu, J., Goswami, V., Rohrbach, M., Parikh, D., and Lee, S. 12-in-1: Multi-task vision and language representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10437–10446, 2020.

Nguyen, D.-K., Goswami, V., and Chen, X. Revisiting modulated convolutions for visual counting and beyond. *arXiv preprint arXiv:2004.11883*, 2020.

Ordonez, V., Kulkarni, G., and Berg, T. Im2text: Describing images using 1 million captioned photographs. *Advances in neural information processing systems*, 24:1143–1151, 2011.

Perez, E., Strub, F., De Vries, H., Dumoulin, V., and Courville, A. Film: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

Plummer, B. A., Wang, L., Cervantes, C. M., Caicedo, J. C., Hockenmaier, J., and Lazebnik, S. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *Proceedings of the IEEE international conference on computer vision*, pp. 2641–2649, 2015.

Qi, D., Su, L., Song, J., Cui, E., Bharti, T., and Sacheti, A. Imagebert: Cross-modal pre-training with large-scale weak-supervised image-text data. *arXiv preprint arXiv:2001.07966*, 2020.

Radford, A., Sutskever, I., Kim, J., Krueger, G., and Agarwal, S. Learning transferable visual models from natural language supervision, 2021.

Ren, S., He, K., Girshick, R., and Sun, J. Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, 39(6):1137–1149, 2016.

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115(3): 211–252, 2015.

Huang, Z., Zeng, Z., Liu, B., Fu, D., and Fu, J. Pixel-bert：通过深度多模态 Transformer 对齐图像像素与文本。*arXiv preprint arXiv:2004.00849*, 2020.

Jiang, H., Misra, I., Rohrbach, M., Learned-Miller, E., 和 Chen, X. 为视觉问答中的网格特征辩护。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 10267–10276 页，2020 年。

Jiang, Y., Natarajan, V., Chen, X., Rohrbach, M., Batra, D., 和 Parikh, D. Pythia v0. 1：2018 年 VQA 挑战赛的获胜参赛作品。*arXiv preprint arXiv:1807.09956*, 2018.

Karpathy, A. 和 Fei-Fei, L. 用于生成图像描述的深度视觉-语义对齐。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 3128–3137 页，2015 年。

Krishna, R., Zhu, Y., Groth, O., Johnson, J., Hata, K., Kravitz, J., Chen, S., Kalantidis, Y., Li, L.-J., Shamma, D. A., 等。Visual Genome：使用众包密集图像标注连接语言与视觉。*In- ternational journal of computer vision*, 123 (1):32–73, 2017.

Krizhevsky, A.、Sutskever, I. 和 Hinton, G. E. 使用深度卷积神经网络进行 Imagenet 分类。见 *NIPS*，2012。

Kuznetsova, A., Rom, H., Alldrin, N., Uijlings, J., Krasin, I., Pont-Tuset, J., Kamali, S., Popov, S., Malloci, M., Kolesnikov, A., 等。Open Images 数据集 v4。*International Journal of Computer Vision*，第 1–26 页，2020。

Lee, K.-H., Chen, X., Hua, G., Hu, H., 和 He, X. 用于图像-文本匹配的堆叠交叉注意力。收录于 *Proceedings of the European Conference on Computer Vision (ECCV)*，第 201–216 页，2018。

Li, G., Duan, N., Fang, Y., Gong, M., Jiang, D., 和 Zhou, M. Unicoder-vl：通过跨模态预训练实现的视觉与语言通用编码器。发表于 *AAAI*，第 11336–11344 页，2020a。

Li, L. H.、Yatskar, M.、Yin, D.、Hsieh, C.-J. 和 Chang, K.-W. VisualBERT：一种用于视觉与语言的简单且高性能的基线。*arXiv preprint arXiv:1908.03557*，2019。

Li, X., Yin, X., Li, C., Zhang, P., Hu, X., Zhang, L., Wang, L., Hu, H., Dong, L., Wei, F., 等。Oscar：面向视觉-语言任务的对象语义对齐预训练。载于 *European Conference on Computer Vision*，第 121–137 页。Springer，2020b。

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P. 和 Zitnick, C. L. Microsoft COCO：上下文中的常见对象。载于 *European conference on computer vision*，第 740–755 页。Springer，2014。

Loshchilov, I. 和 Hutter, F. 解耦权重衰减正则化。见 *International Conference on Learning Representations*，2018。

Lu, J., Batra, D., Parikh, D., 和 Lee, S. Vilbert：为视觉与语言任务预训练任务无关的视觉语言表征。载于 *Advances in Neural Information Processing Systems*，第 13–23 页，2019。

Lu, J., Goswami, V., Rohrbach, M., Parikh, D., 和 Lee, S. 12-in-1：多任务视觉与语言表征学习。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 10437–10446 页，2020。

Nguyen, D.-K.、Goswami, V. 和 Chen, X. 重新审视用于视觉计数及其他任务的调制卷积。*arXiv preprint arXiv:2004.11883*, 2020.

Ordonez, V., Kulkarni, G., 和 Berg, T. Im2text：使用100万张带说明文字的照片描述图像。*Advances in neural information processing systems*, 24:1143–1151, 2011.

Perez, E.、Strub, F.、De Vries, H.、Dumoulin, V. 和 Courville, A. FiLM：使用通用条件层进行视觉推理。载于 *Proceedings of the AAAI Conference on Artificial Intelligence*，第 32 卷，2018 年。

Plummer, B. A., Wang, L., Cervantes, C. M., Caicedo, J. C., Hockenmaier, J., 和 Lazebnik, S. Flickr30k 实体：为更丰富的图像到句子模型收集区域到短语的对应关系。见 *Proceedings of the IEEE international conference on computer vision*，第 2641–2649 页，2015。

Qi, D., Su, L., Song, J., Cui, E., Bharti, T., 和 Sacheti, A. Imagebert：使用大规模弱监督图像-文本数据进行跨模态预训练。*arXiv preprint arXiv:2001.07966*, 2020.

Radford, A.、Sutskever, I.、Kim, J.、Krueger, G. 和 Agarwal, S. 通过自然语言监督学习可迁移的视觉模型，2021年。

Ren, S.、He, K.、Girshick, R. 和 Sun, J. Faster r-cnn：通过区域候选网络实现实时目标检测。*IEEE transactions on pattern analysis and machine intelligence*，39(6):1137–1149，2016。

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., 等。Imagenet 大规模视觉识别挑战。*International journal of computer vision*, 115(3): 211–252, 2015.

Sharma, P., Ding, N., Goodman, S., and Soricut, R. Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2556–2565, 2018.

Shorten, C. and Khoshgoftaar, T. M. A survey on image data augmentation for deep learning. *Journal of Big Data*, 6(1):1–48, 2019.

Su, W., Zhu, X., Cao, Y., Li, B., Lu, L., Wei, F., and Dai, J. Vl-bert: Pre-training of generic visual-linguistic representations. *arXiv preprint arXiv:1908.08530*, 2019.

Suhr, A., Zhou, S., Zhang, A., Zhang, I., Bai, H., and Artzi, Y. A corpus for reasoning about natural language grounded in photographs. *arXiv preprint arXiv:1811.00491*, 2018.

Tan, H. and Bansal, M. Lxmert: Learning cross-modality encoder representations from transformers. *arXiv preprint arXiv:1908.07490*, 2019.

Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., and Jégou, H. Training data-efficient image transformers & distillation through attention. *arXiv preprint arXiv:2012.12877*, 2020.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. *Advances in neural information processing systems*, 30:5998–6008, 2017.

Xie, S., Girshick, R., Dollár, P., Tu, Z., and He, K. Aggregated residual transformations for deep neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1492–1500, 2017.

Xie, Y., Wang, X., Wang, R., and Zha, H. A fast proximal point method for computing exact wasserstein distance. In *Uncertainty in Artificial Intelligence*, pp. 433–453. PMLR, 2020.

Yu, F., Tang, J., Yin, W., Sun, Y., Tian, H., Wu, H., and Wang, H. Ernie-vil: Knowledge enhanced vision-language representations through scene graph. *arXiv preprint arXiv:2006.16934*, 2020.

Yu, Z., Yu, J., Cui, Y., Tao, D., and Tian, Q. Deep modular co-attention networks for visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6281–6290, 2019.

Yun, S., Han, D., Oh, S. J., Chun, S., Choe, J., and Yoo, Y. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 6023–6032, 2019.

Zhang, H., Cisse, M., Dauphin, Y. N., and Lopez-Paz, D. mixup: Beyond empirical risk minimization. *arXiv preprint arXiv:1710.09412*, 2017.

Zhang, P., Li, X., Hu, X., Yang, J., Zhang, L., Wang, L., Choi, Y., and Gao, J. Vinvl: Making visual representations matter in vision-language models. *arXiv preprint arXiv:2101.00529*, 2021.

Zhou, L., Palangi, H., Zhang, L., Hu, H., Corso, J. J., and Gao, J. Unified vision-language pre-training for image captioning and vqa. In *AAAI*, pp. 13041–13049, 2020.

Sharma, P., Ding, N., Goodman, S., and Soricut, R. 概念字幕：用于自动图像字幕生成的经过清洗、上位词化的图像替代文本数据集。载于 *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*，第 2556–2565 页，2018 年。

Shorten, C. 和 Khoshgoftaar, T. M. 用于深度学习的图像数据增强综述。*Journal of Big Data*, 6(1):1–48, 2019.

Su, W., Zhu, X., Cao, Y., Li, B., Lu, L., Wei, F., 和 Dai, J. Vl-bert：通用视觉-语言表示的预训练。*arXiv preprint arXiv:1908.08530*, 2019.

Suhr, A., Zhou, S., Zhang, A., Zhang, I., Bai, H., and Artzi, Y. 一个用于基于照片进行自然语言推理的语料库。*arXiv preprint arXiv:1811.00491*, 2018.

Tan, H. 和 Bansal, M. Lxmert：从 Transformer 学习跨模态编码器表示。*arXiv preprint arXiv:1908.07490*，2019。

Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., and Jégou, H. 训练数据高效的图像 Transformer 与通过注意力进行蒸馏。*arXiv preprint arXiv:2012.12877*, 2020.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, ., 和 Polosukhin, I. 注意力就是你所需要的一切。*Advances in neural information processing systems*, 30:5998–6008, 2017.

Xie, S., Girshick, R., Dollár, P., Tu, Z., 和 He, K. 用于深度神经网络的聚合残差变换。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 1492–1500 页，2017 年。

谢，Y.，王，X.，王，R.，和查，H. 一种用于计算精确 Wasserstein 距离的快速近端点方法。见 *Uncertainty in Artificial Intelligence*，第 433–453 页。PMLR，2020。

Yu, F., Tang, J., Yin, W., Sun, Y., Tian, H., Wu, H., and Wang, H. Ernie-vil：通过场景图增强知识的视觉-语言表示。*arXiv preprint arXiv:2006.16934*, 2020.

Yu, Z.、Yu, J.、Cui, Y.、Tao, D. 和 Tian, Q. 用于视觉问答的深度模块化协同注意力网络。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 6281–6290 页，2019 年。

Yun, S., Han, D., Oh, S. J., Chun, S., Choe, J., 和 Yoo, Y. Cutmix：训练具有可定位特征的强分类器的正则化策略。载于 *Proceedings of the IEEE/CVF International Conference on Computer Vision*，第 6023–6032 页，2019。

Zhang, H., Cisse, M., Dauphin, Y. N., and Lopez-Paz, D. mixup：超越经验风险最小化。*arXiv preprint arXiv:1710.09412*，2017。

Zhang, P., Li, X., Hu, X., Yang, J., Zhang, L., Wang, L., Choi, Y., 和 Gao, J. VinVL：让视觉表征在视觉-语言模型中发挥作用。*arXiv preprint arXiv:2101.00529*, 2021.

Zhou, L., Palangi, H., Zhang, L., Hu, H., Corso, J. J., and Gao, J. 用于图像描述和 VQA 的统一视觉-语言预训练。载于 *AAAI*，第 13041–13049 页，2020。